

Anatomy of The Ear

Lesson topic: The Ear

Time: 2hrs

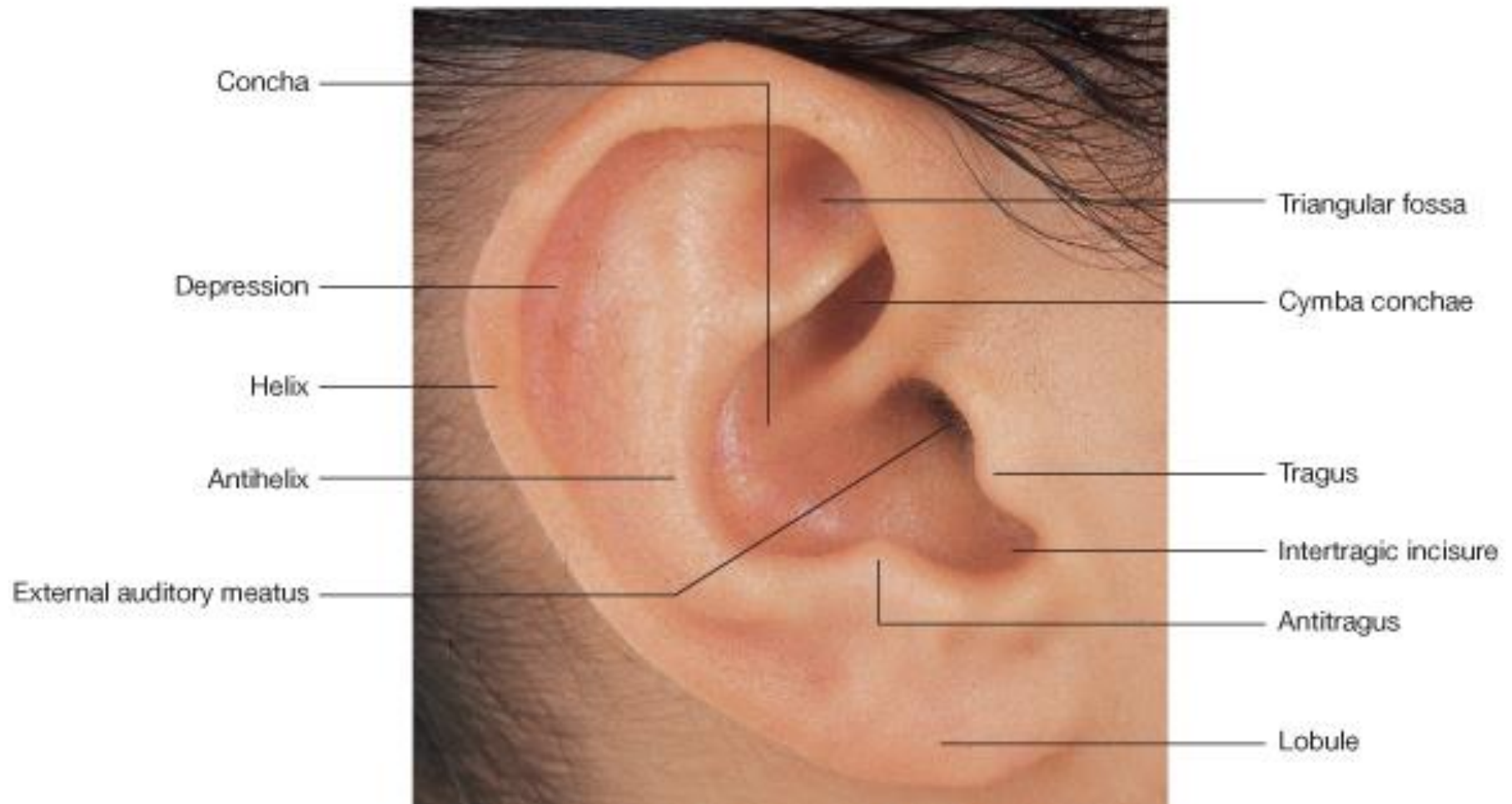
■ Outline:

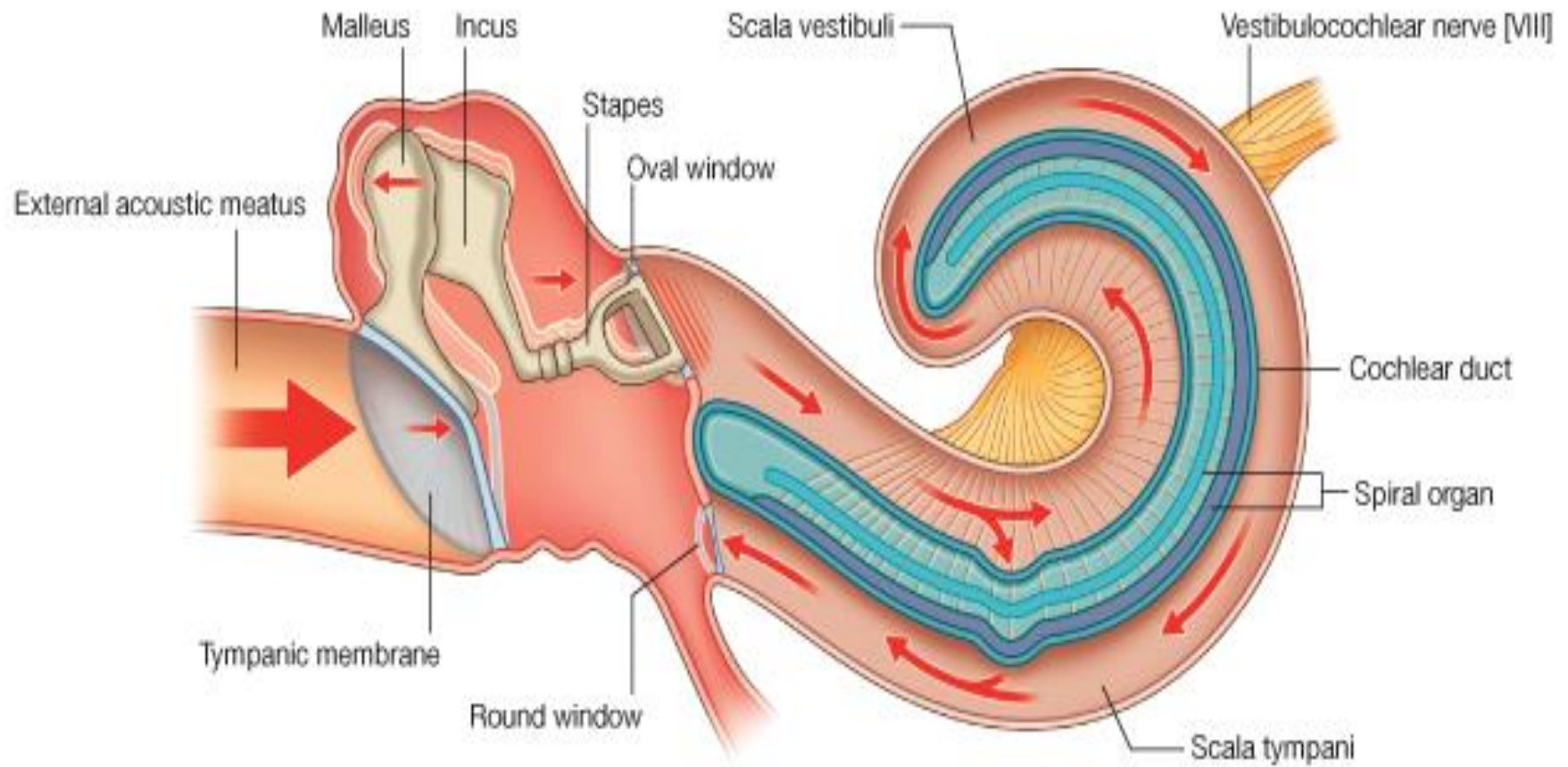
- External Ear
- Middle ear
- Inner ear
- Nonvascular structures of the ear

Objectives:

- At the end of this lesson students will be able to:
 - Define the ear anatomy
 - Describe the three parts of the ear
 - Enumerate the walls of the middle ear
 - Demonstrate the contents of the middle and inner ear
 - Define the communications of the middle and inner ear

THE EAR



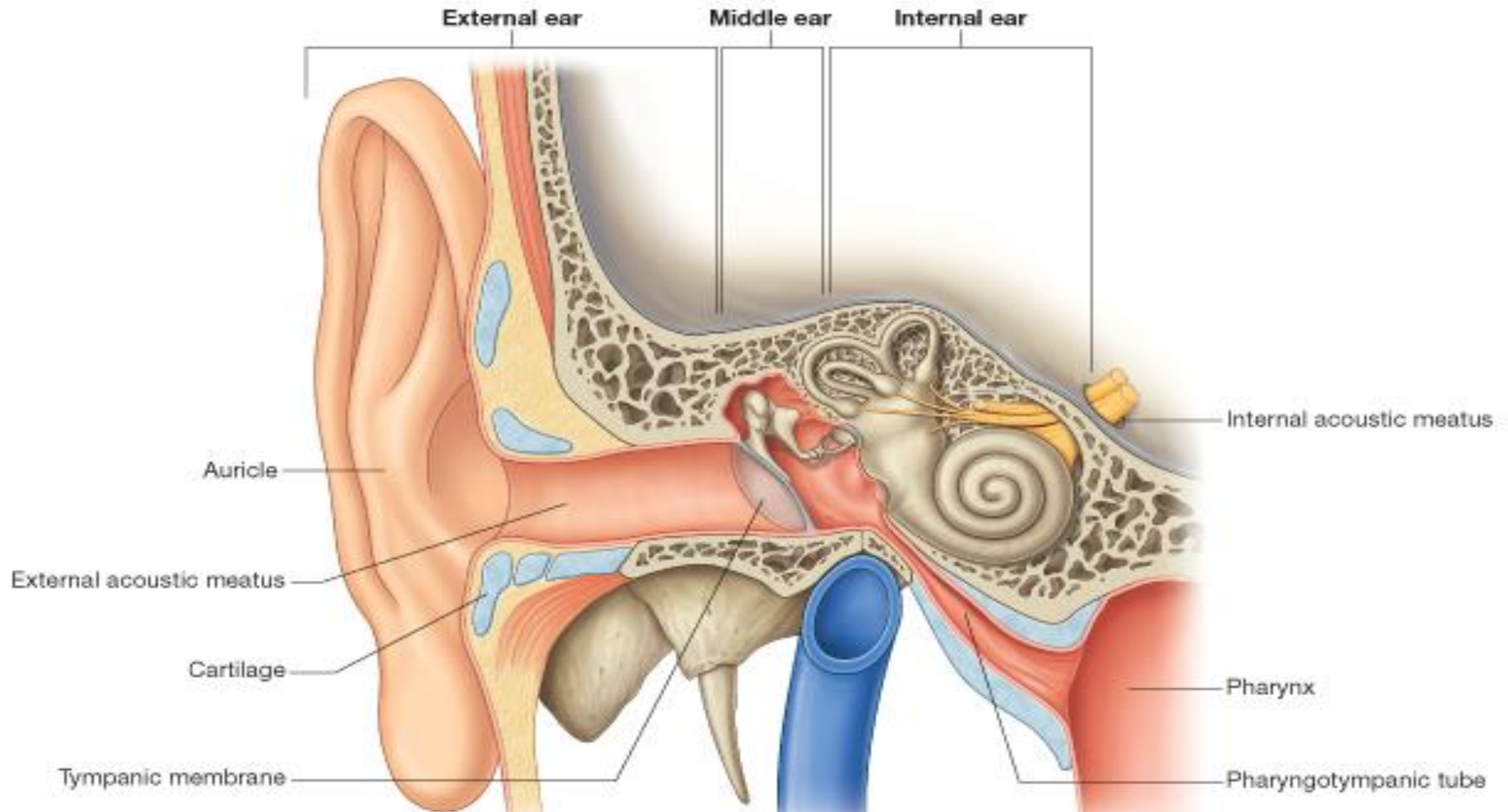


EAR

- is a complex and delicate organ of hearing
- collects sound waves so one can hear the world around
- has a second function—it helps keep balance
- has three parts:
 - the external ear:
 - consisting of:
 - the part attached to the lateral aspect of the head
 - the canal leading inward
 - the middle ear:
 - Air filled cavity in the petrous part of the temporal bone
 - bounded laterally, and separated from the external canal by a membrane
 - connected internally to the naso pharynx by a narrow tube

■ The internal ear:

- consisting of a series of cavities within the petrous part of the temporal bone between the middle ear laterally and the internal acoustic meatus medially.



■ External ear

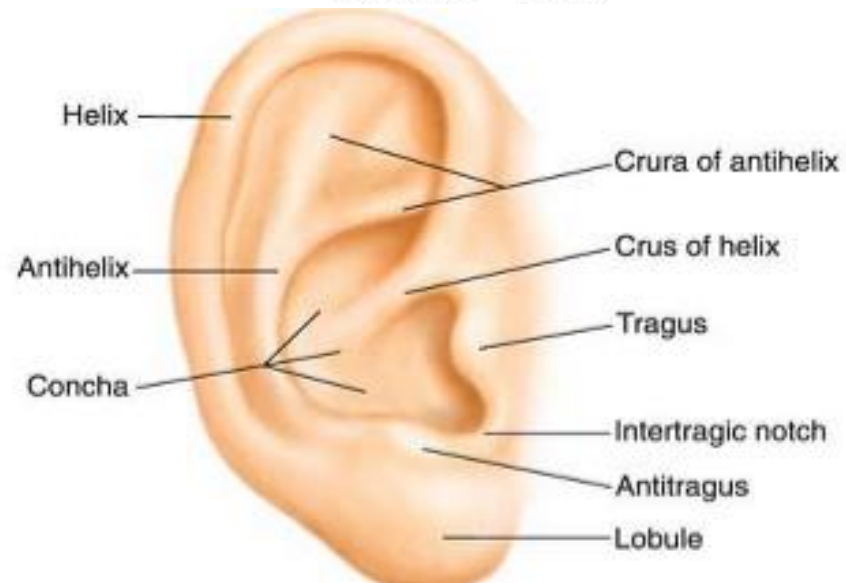
– Consists of two parts

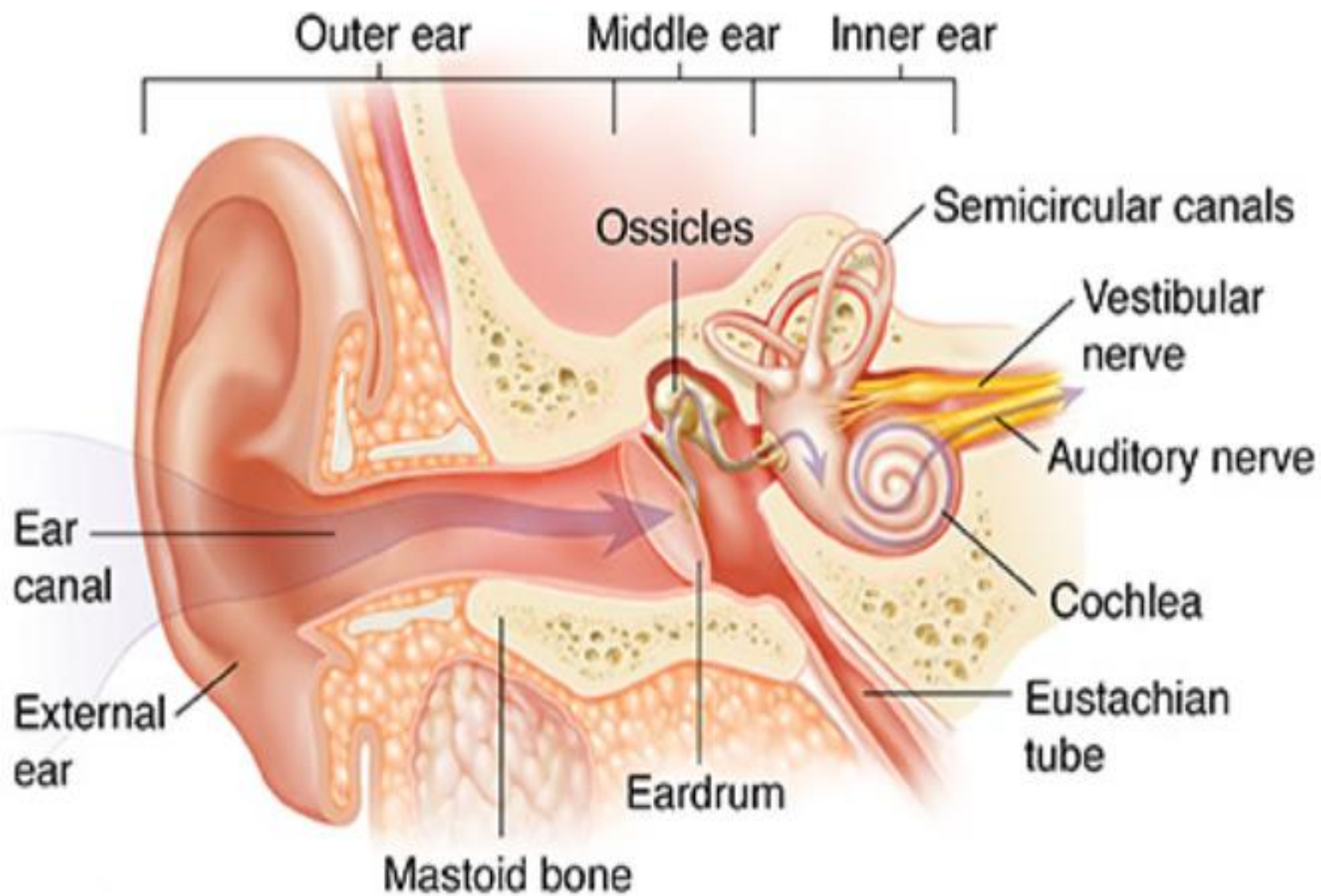
- The part projecting from the side of the head is **the auricle (pinna)**
- the canal leading inward is **the external acoustic meatus**

1. Auricle(Pinna)

- is on the side of the head and assists in capturing sound
- consists of cartilage covered with skin and arranged in a pattern of various elevations and depressions
- The large outside rim of the auricle is **the helix**
- It ends inferiorly at the **fleshy lobule**, the only part of the auricle **not supported by cartilage**

- The hollow center of the auricle is the **concha of auricle**
- Just anterior to the opening of the external acoustic meatus, in front of the concha, is an elevation (**the tragus**)
- Opposite the tragus, and above the fleshy lobule, is another elevation (**the antitragus**)
- A smaller curved rim, parallel and anterior to the helix is the antihelix





Skin

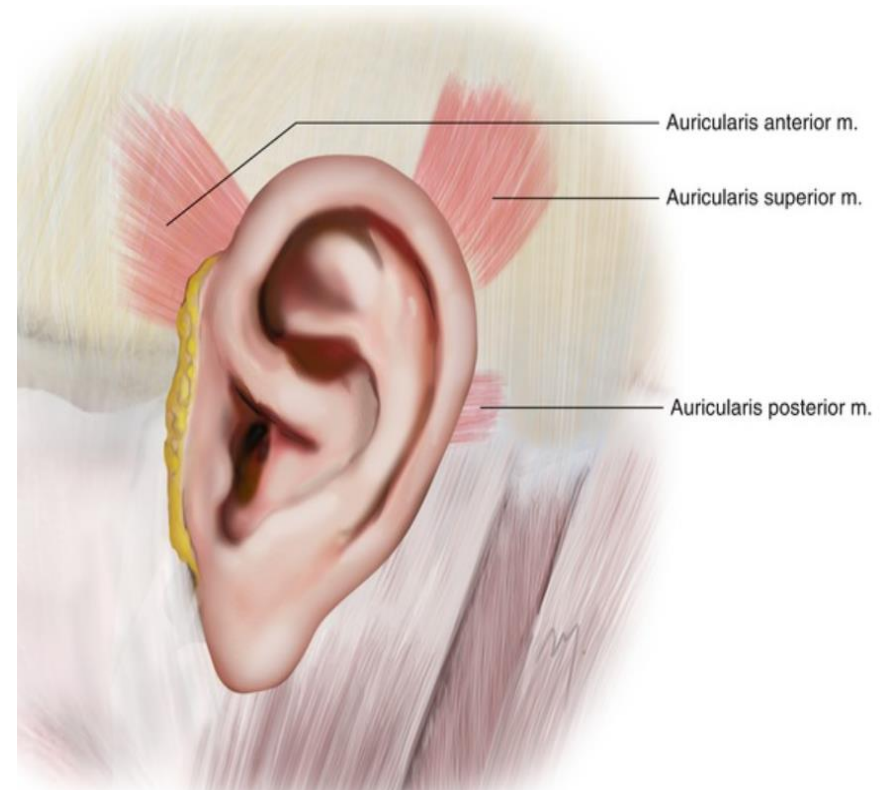
- Thin with no dermal papillae
- Closely adherent to underlying cartilage & bony wall
- Cartilagenous part has subcutaneous tissue -secretes wax
- Ceruminous glands and hair follicles are limited to cartilagenous parts only

Ligaments

- **Extrinsic**
 - connects auricle to temporal bone
 - **Anterior ligament** – tragus & spine of helix to root of zygomatic process
 - **Posterior ligament** – post surf. of concha to lat. surf of mastoid process
- **Intrinsic**
 - connects individual auricular cartilages
 - Strong fibrous band between tragus and helix
 - Another band between antihelix and tail of helix

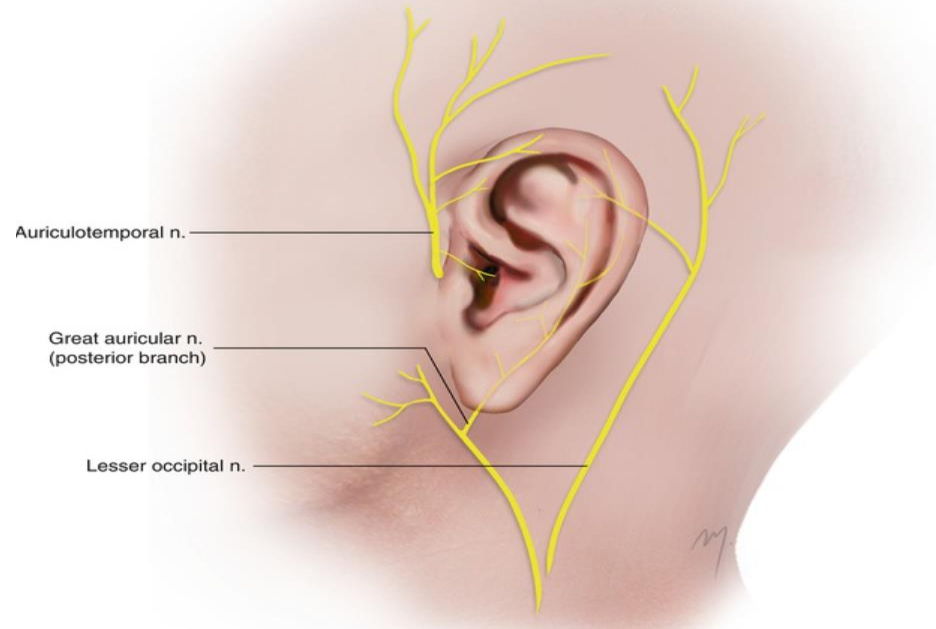
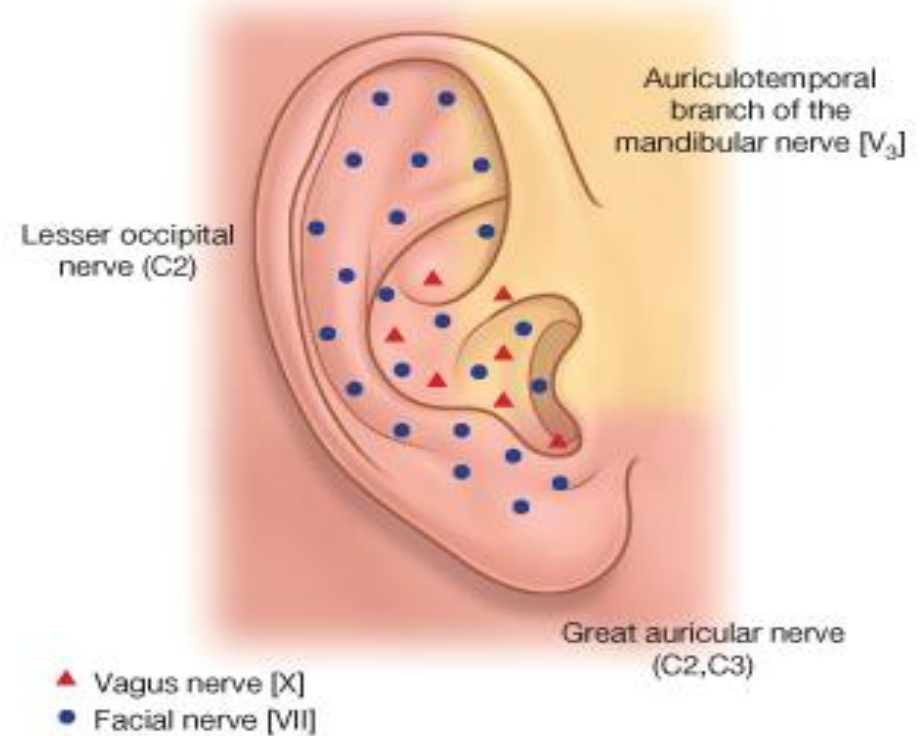
Muscles

- **Intrinsic** and **extrinsic** muscles
- **The intrinsic muscles:**
 - pass between the cartilaginous parts of the auricle
 - may change the shape of the auricle;
- **The extrinsic muscles:**
 - the anterior, superior, and posterior auricular muscles
 - pass from the scalp or skull to the auricle
 - and may also play a role in positioning of the auricle.



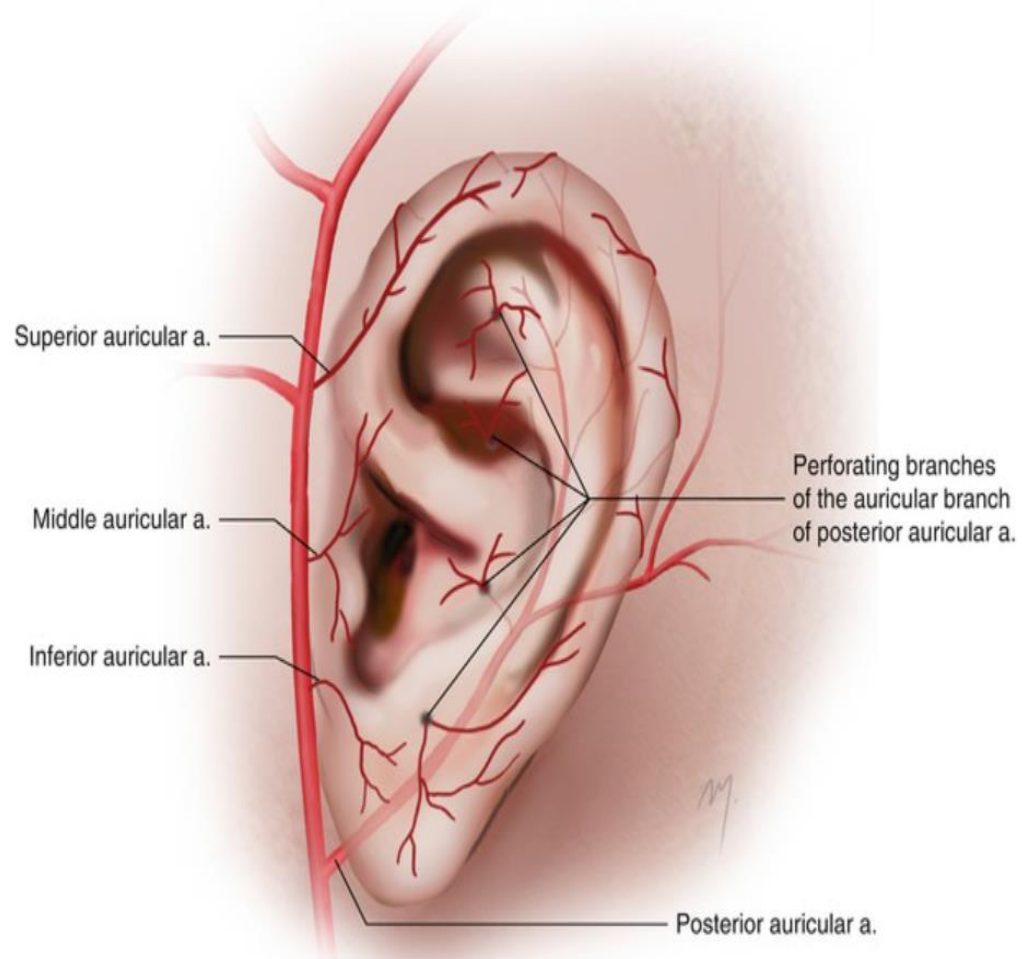
Sensory Nerve supply

- from many sources:
- outer more superficial surfaces are supplied by:
 - ✓ the great auricular nerve
 - ✓ and lesser occipital nerves from the cervical plexus
 - ✓ the auriculotemporal branch of the mandibular nerve [V3].
- deeper parts are supplied by:
 - ✓ branches from the facial nerve [VII]
 - ✓ the vagus nerve [X].



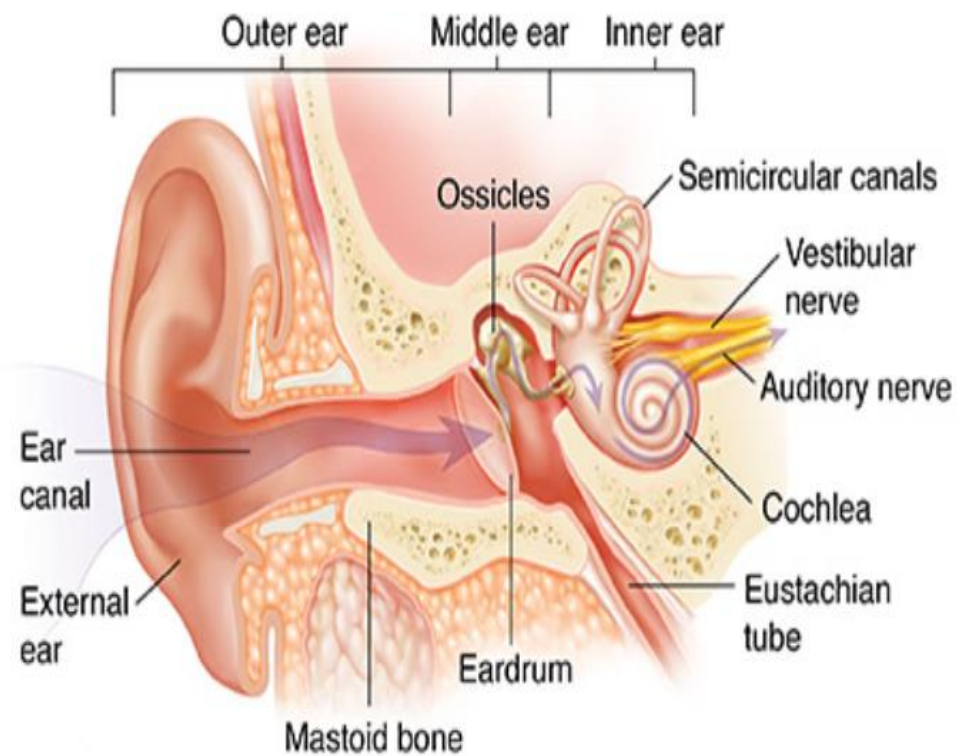
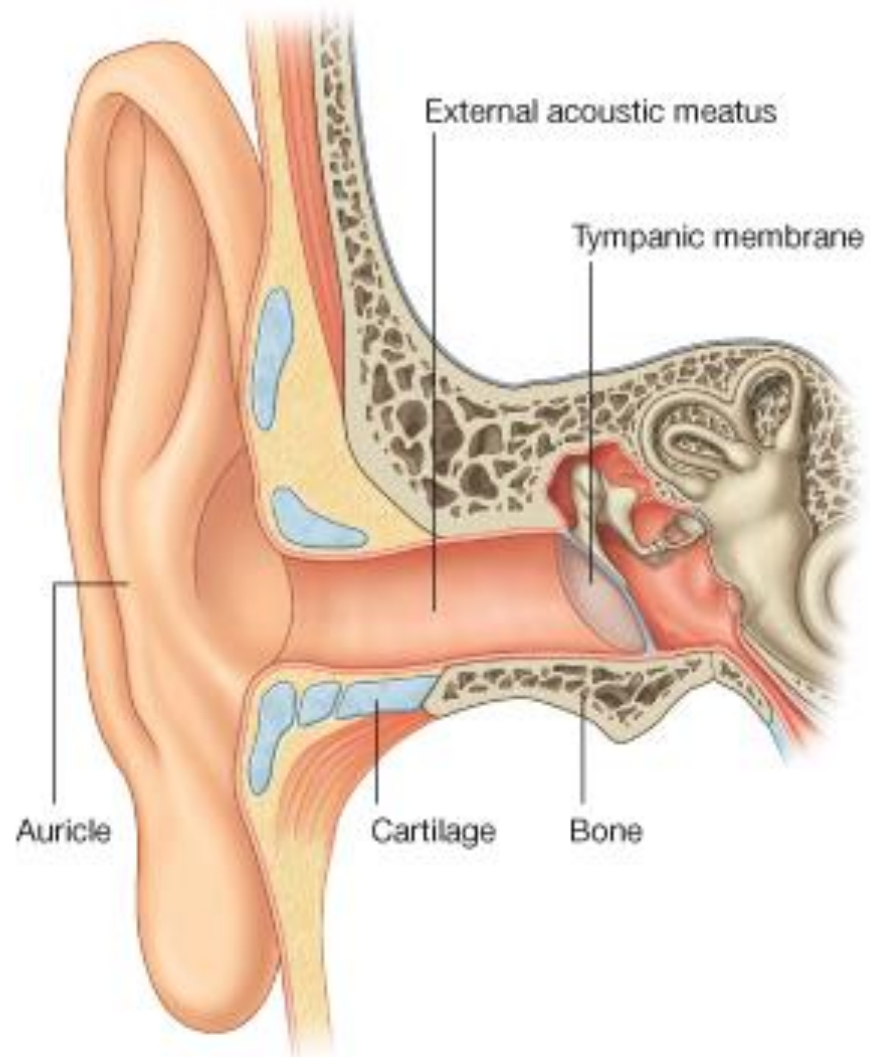
Blood supply

- The arterial supply from:
 - ✓ posterior auricular artery a
 - ✓ the superficial temporal artery
- Venous drainage
 - ✓ is through vessels following the arteries.
- Lymphatic drainage:
 - ✓ anteriorly into parotid nodes
 - ✓ posteriorly into mastoid nodes
 - ✓ and possibly into the upper deep cervical nodes.



2. External acoustic meatus

- ✓ extends from the deepest part of the concha to the tympanic membrane (eardrum)
- ✓ **length $\approx$ 1 inch (2.5 cm)**
- ✓ The lateral **one-third** is formed from cartilages
- ✓ The medial **two-thirds** is a bony tunnel in the temporal bone
- ✓ is covered with skin, some of which contains hair and modified **sweat glands** producing cerumen (earwax).
- Its diameter varies: wider laterally and narrow medially.
- **Course:**
 - ✓ passes upwards in an anterior direction
 - ✓ then turns slightly posteriorly still passing in an upwards direction
 - ✓ and finally, turns again in an anterior direction with a slight descent.

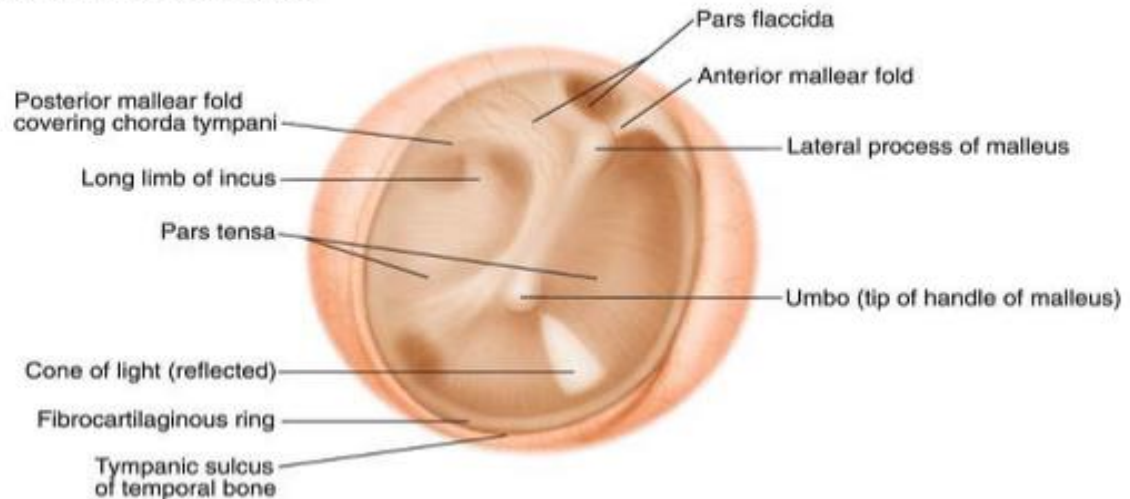


- **Sensory innervation:**
- is from several of the cranial nerves.
- **The major sensory from:**
 - ✓ a branches of the auriculotemporal nerve
 - a branch of the mandibular nerve [V3],
 - ✓ the auricular branch of the vagus nerve [X].
- **Minor sensory inputs**
 - ✓ may travel through branches of the facial nerve [VII].

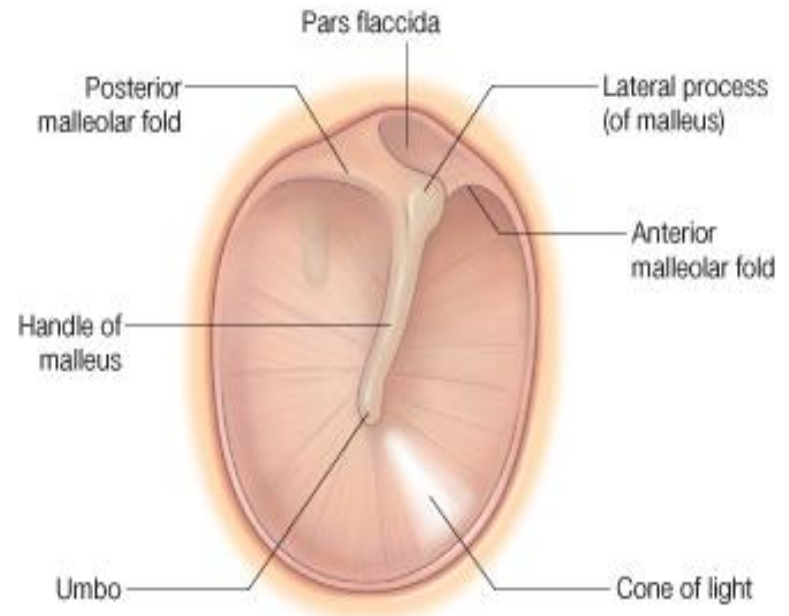
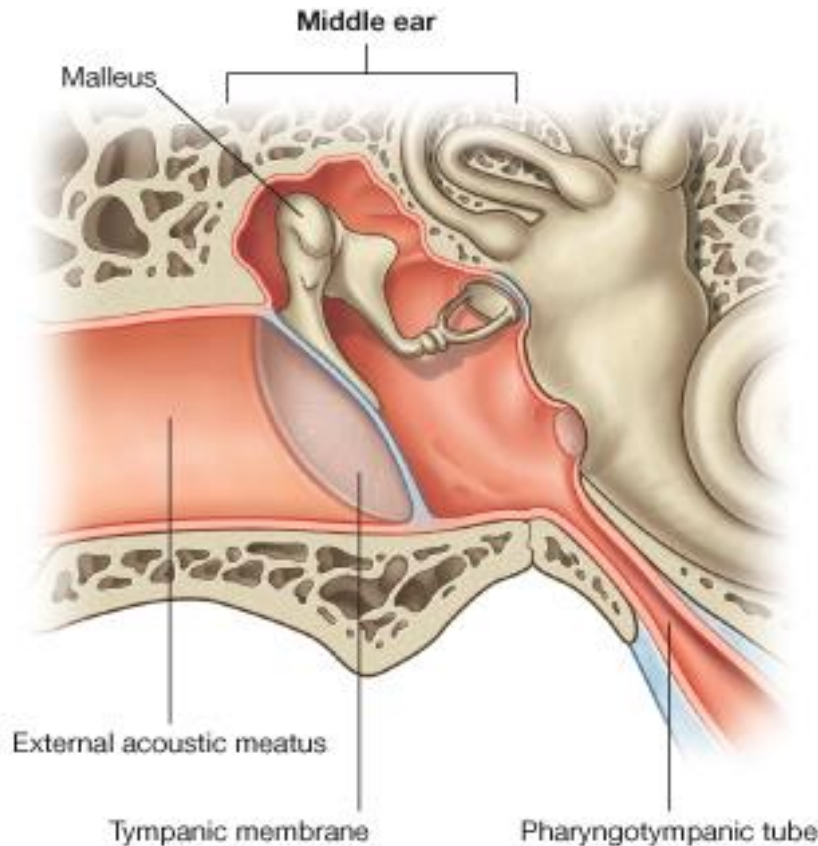
Tympanic membrane

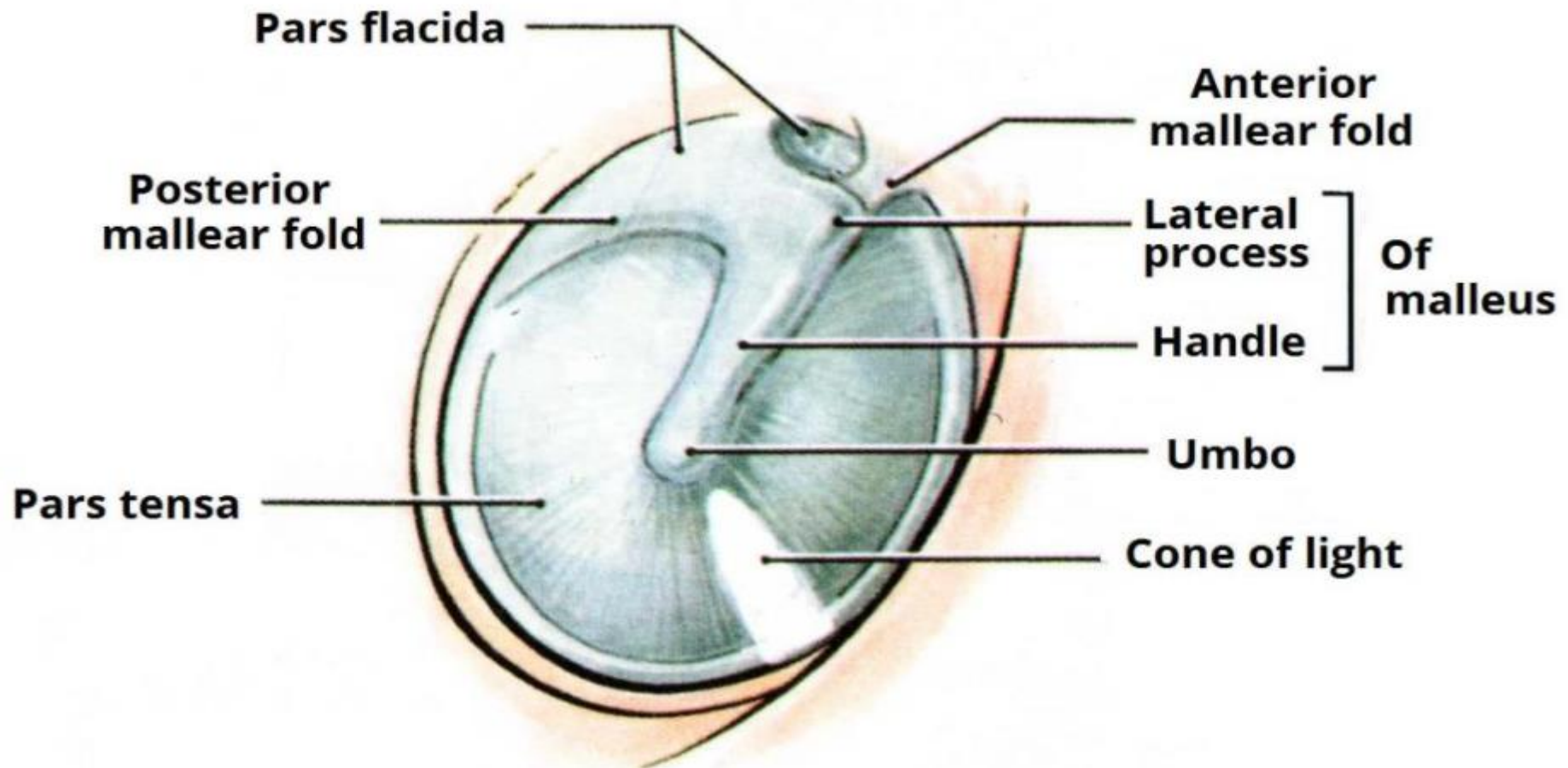
- separates the external acoustic meatus from the middle ear
- It **is at an angle**, sloping medially from top to bottom
- and posteriorly to anteriorly.
- Its lateral surface faces **inferiorly** and **anteriorly**.
- Has a connective tissue core lined with skin on the outside and mucous membrane on the inside.
- Has fibrocartilaginous ring attaches it periphery to the tympanic part of the temporal bone.

Tympanic membrane, lateral view

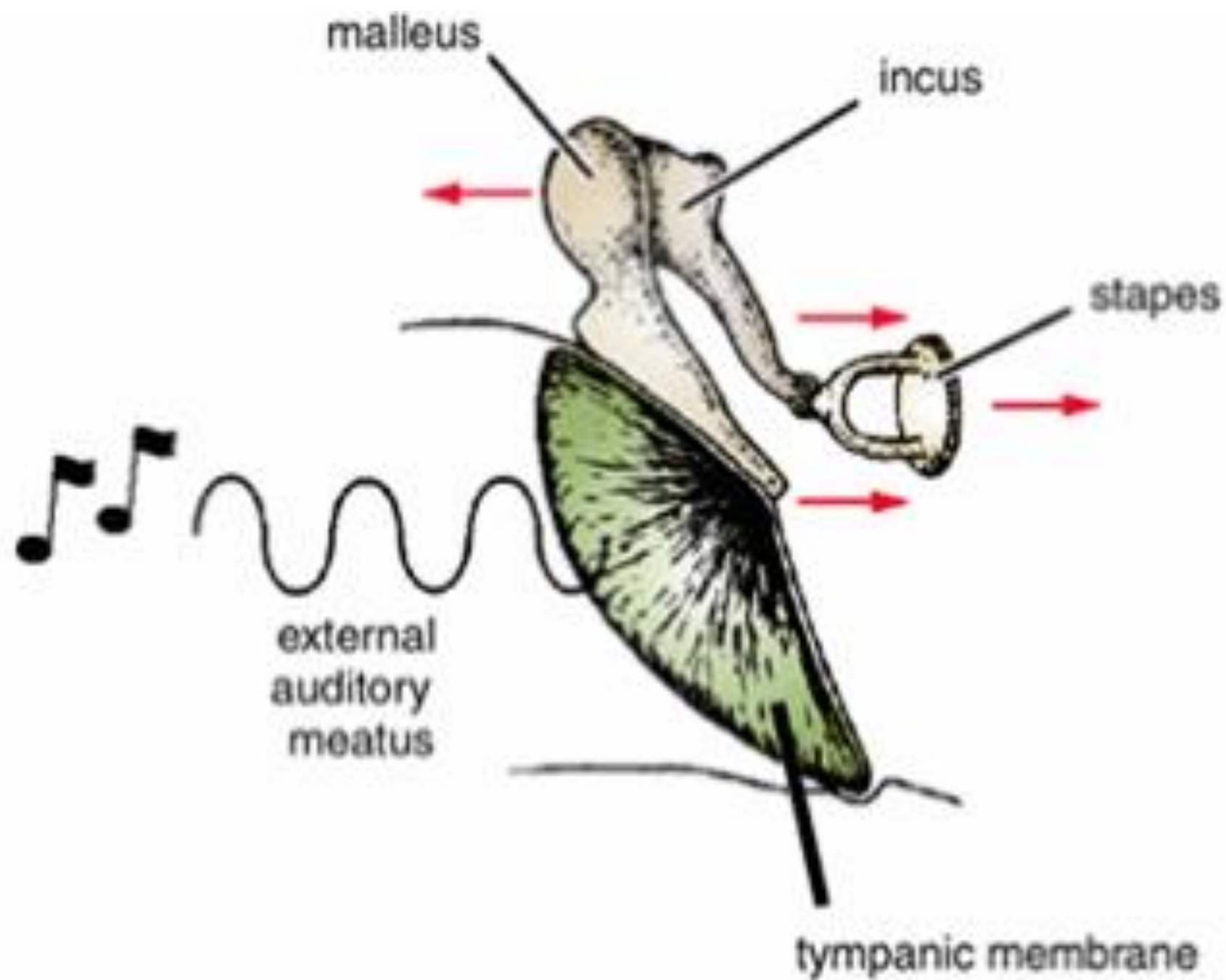


- At its center, a concavity is produced by the attachment on its internal surface of the lower end of the handle of malleus
- This point of attachment is the **umbo of tympanic membrane**.
- * Anteroinferior to the umbo of tympanic membrane a bright reflection of light, referred to **as the cone of light**, is usually visible when examining the tympanic membrane with an **otoscope**.



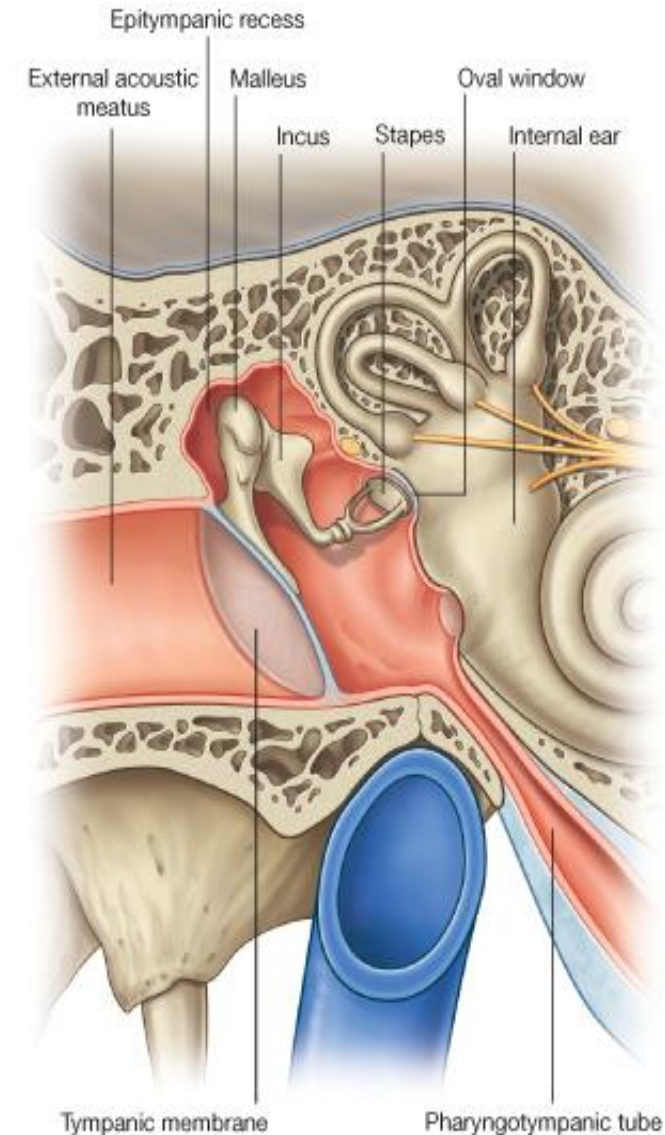


- ✓ The outer surface is primarily by the trigeminal nerve [V] with additional contribution of the facial [VII] and vagus [X] nerves
- ✓ on the inner surface of the tympanic membrane is carried entirely by the glossopharyngeal [IX] nerve.

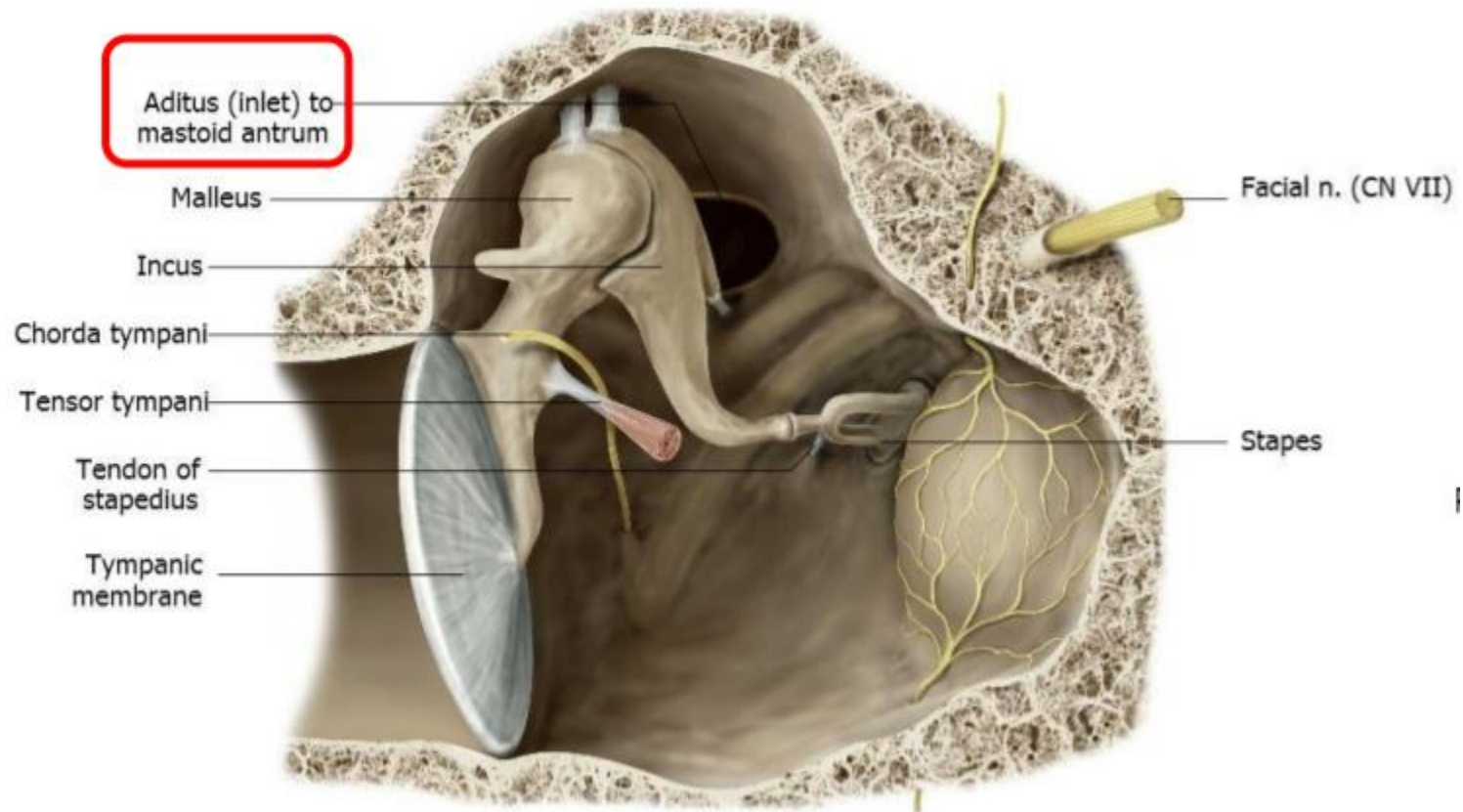


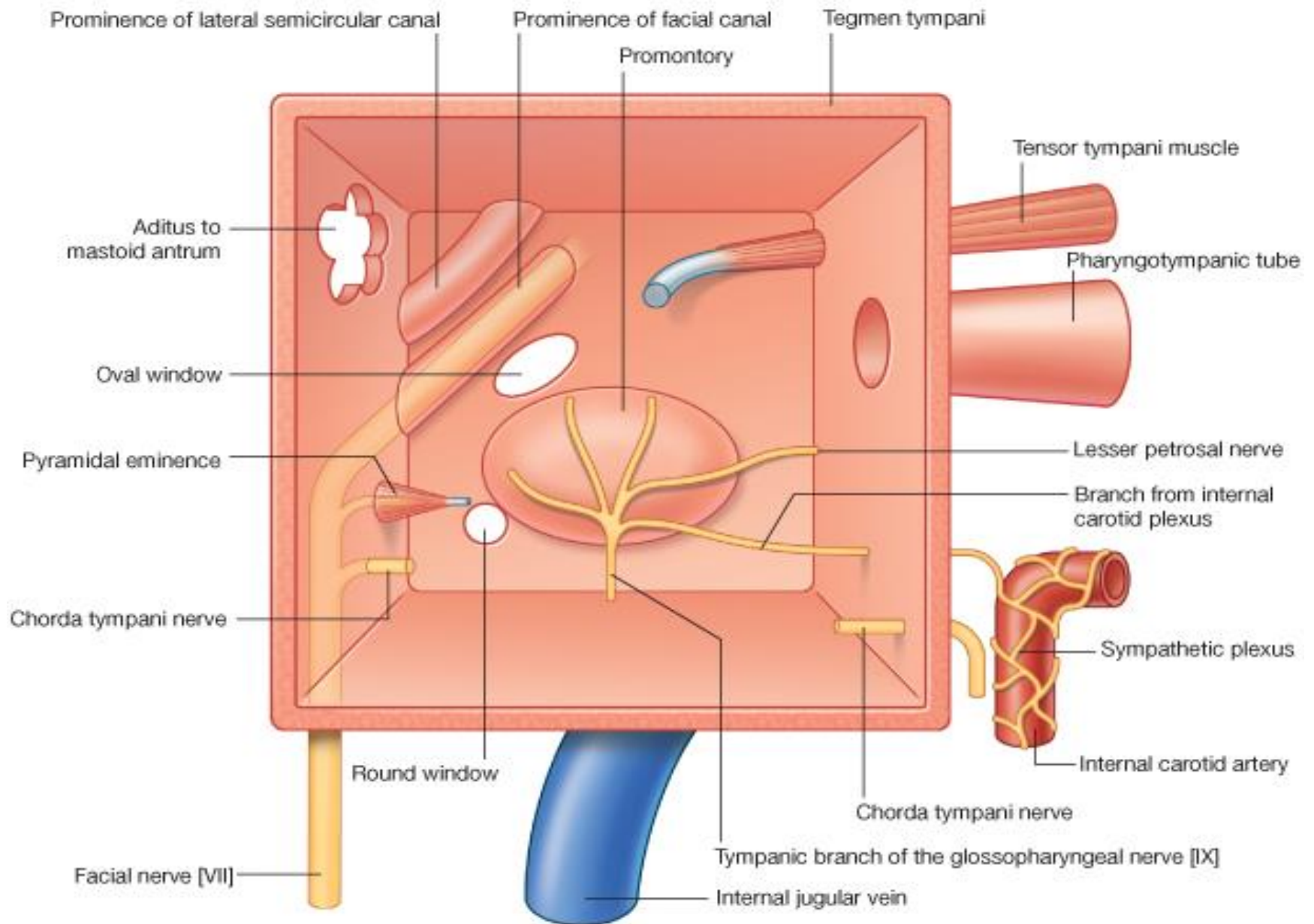
2. Middle ear

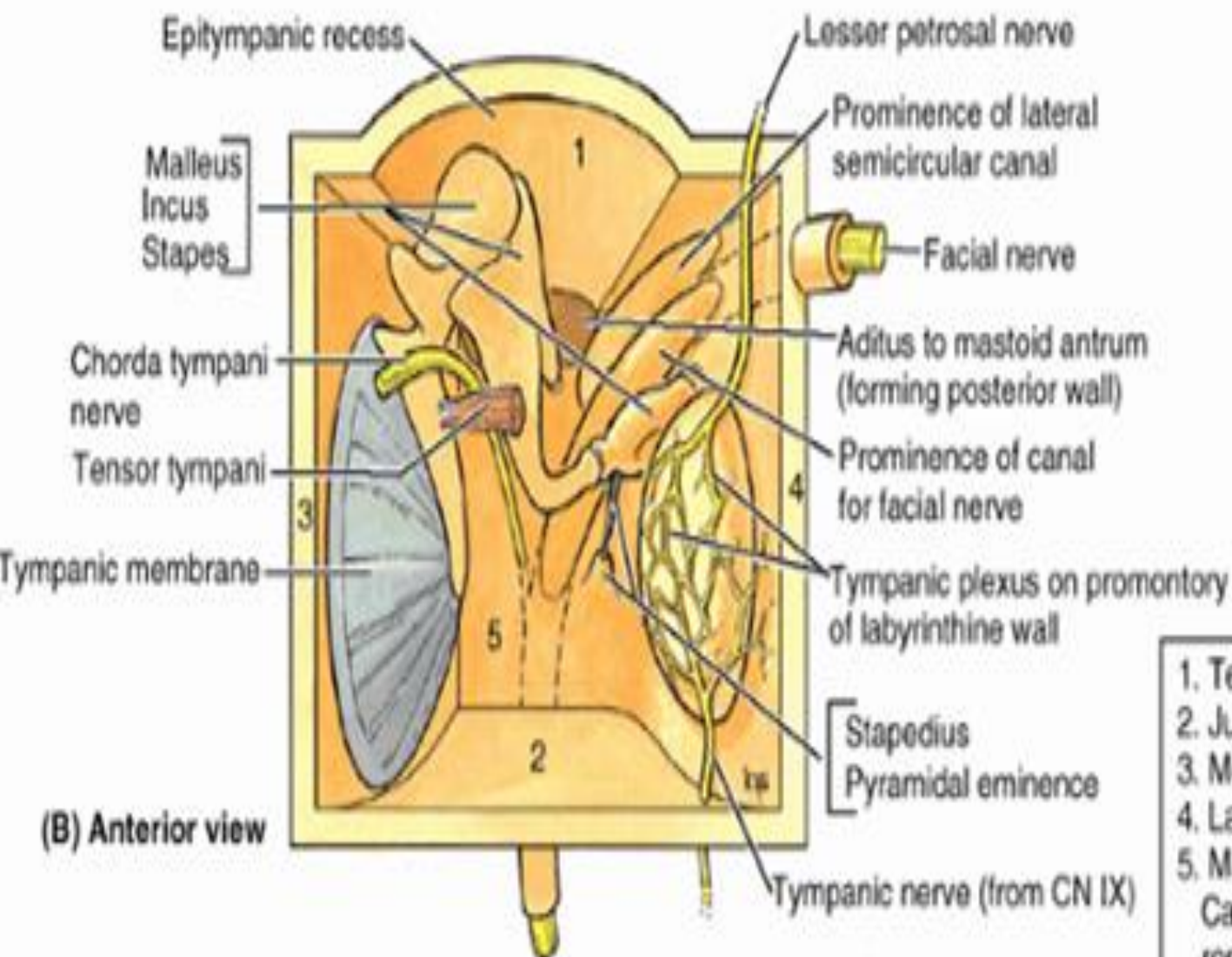
- is an air-filled, mucous membrane-lined space in the temporal bone
- **Location:**
 - between the tympanic membrane **laterally**
 - and the lateral wall of the internal ear **medially**
- **consisting of two parts**
 - the **tympanic cavity** immediately adjacent **to the tympanic membrane**
 - the **epitympanic recess** superiorly
- **Communication:**
 - with the mastoid area posteriorly
 - With the nasopharynx (via the pharyngotympanic tube) anteriorly



Tempanic cavity







1. Tegmental wall (roof)
 2. Jugular wall (floor)
 3. Membranous (lateral) wall
 4. Labyrinthine (medial) wall
 5. Mastoid (posterior) wall
- Carotid (anterior) wall was removed to provide this view

Boundaries of the middle ear

- has six boundaries:
 - a roof and a floor, and anterior, posterior, medial, and lateral walls

1. Tegmental wall (Roof):

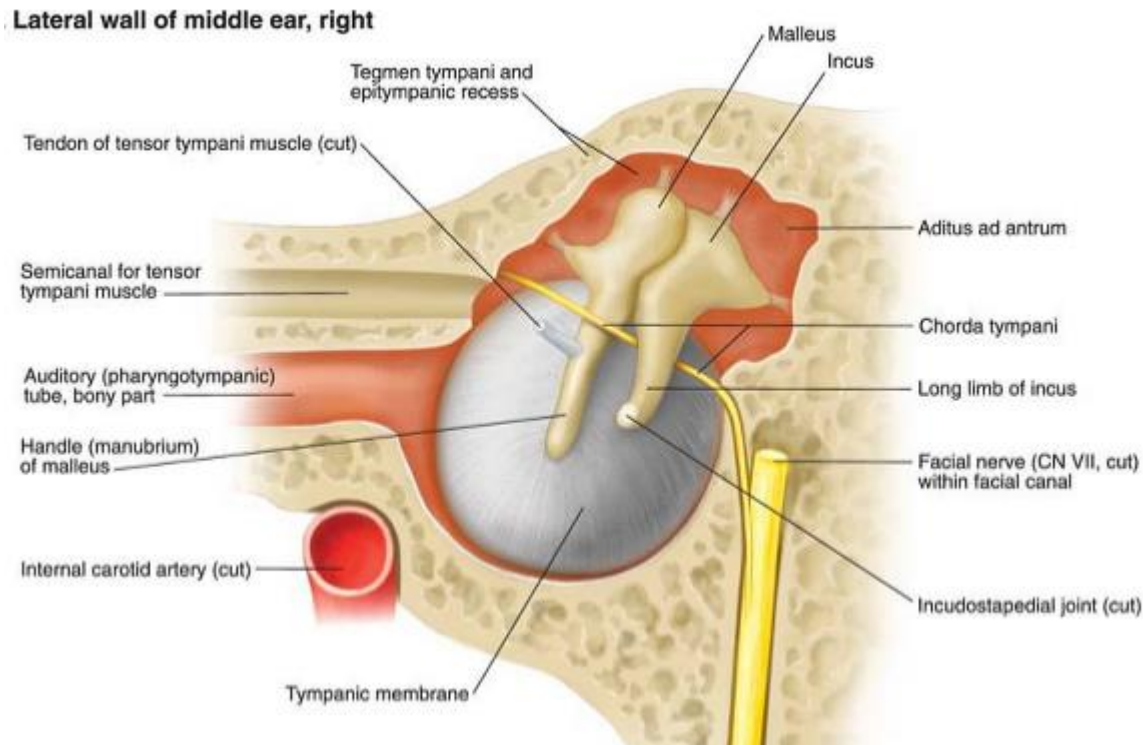
- consists of a thin layer of bone, which separates the middle ear from the middle cranial fossa (tegmen tympani, on the anterior surface of the petrous part of the temporal bone)

2. Jugular wall (Floor):

- consists of a thin layer of bone that separates it from the internal jugular vein
- Occasionally, the floor is thickened by the presence of mastoid air cells
- Near the medial border of the floor is a small aperture, through which the tympanic branch from the glossopharyngeal nerve [IX] enters the middle ear.

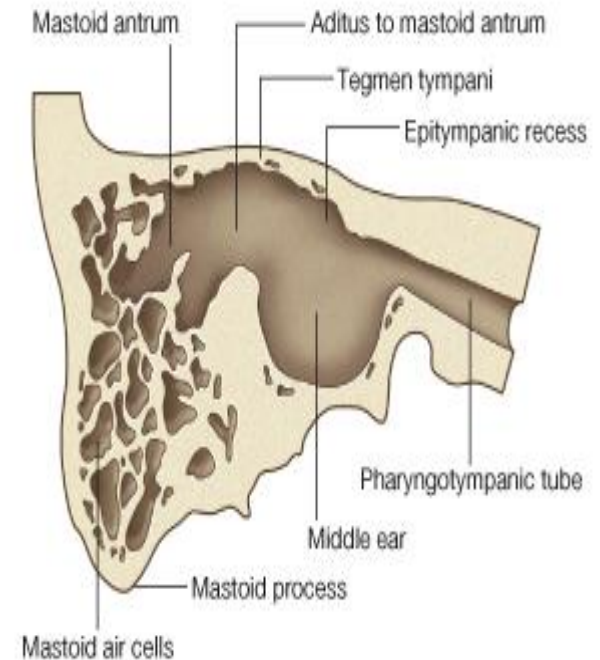
3. Membranous wall (Lateral)

- consists almost entirely of the tympanic membrane
- but because the tympanic membrane does not extend superiorly into the **epitympanic recess**, the upper part of the membranous wall is the bony lateral wall of the **epitympanic recess**.

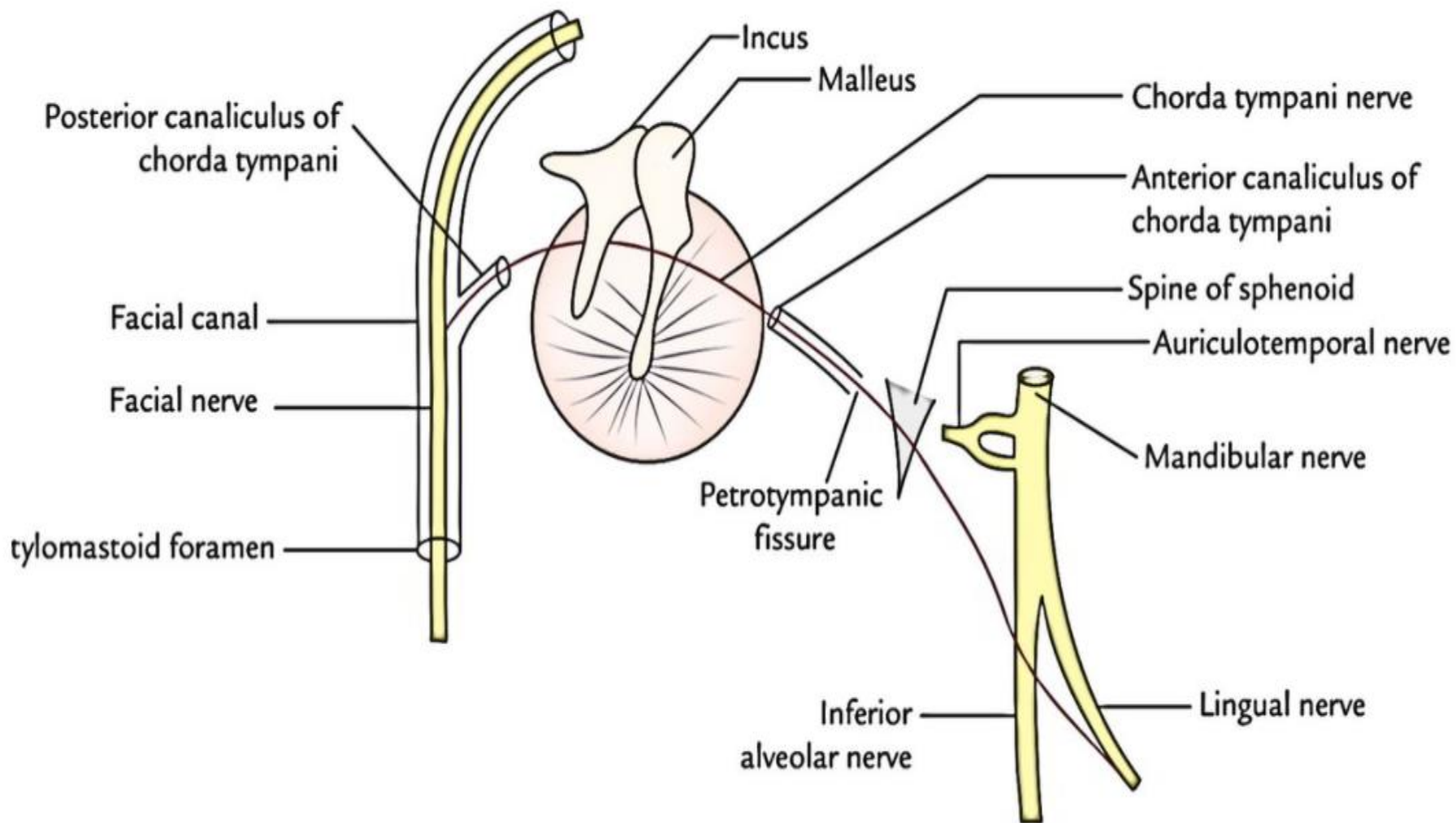


4. Mastoid wall (posterior)

- is only partially complete
- The lower part:
 - ✓ consists of a bony partition b/n **tympenic cavity** and **mastoid air cells**.
- Superiorly, the **epitympanic recess** is continuous with the aditus to the mastoid antrum



- Associated with the mastoid wall are:
 - **the pyramidal eminence**, a small elevation through which the tendon of the **stapedius muscle** enters the middle ear;
 - the opening through which the **chorda tympani nerve**, a branch of the facial nerve [VII], enters the middle ear
- The chorda tympani:
 - is a nerve that arises from the mastoid segment of the facial nerve
 - Due to its intimate relationship to the middle ear (tympanum) it is called chorda tympani nerve
 - Carries afferent special sensation from the anterior two-thirds of the tongue via the lingual nerve
 - As well as efferent parasympathetic secretomotor innervation to the submandibular and sublingual glands

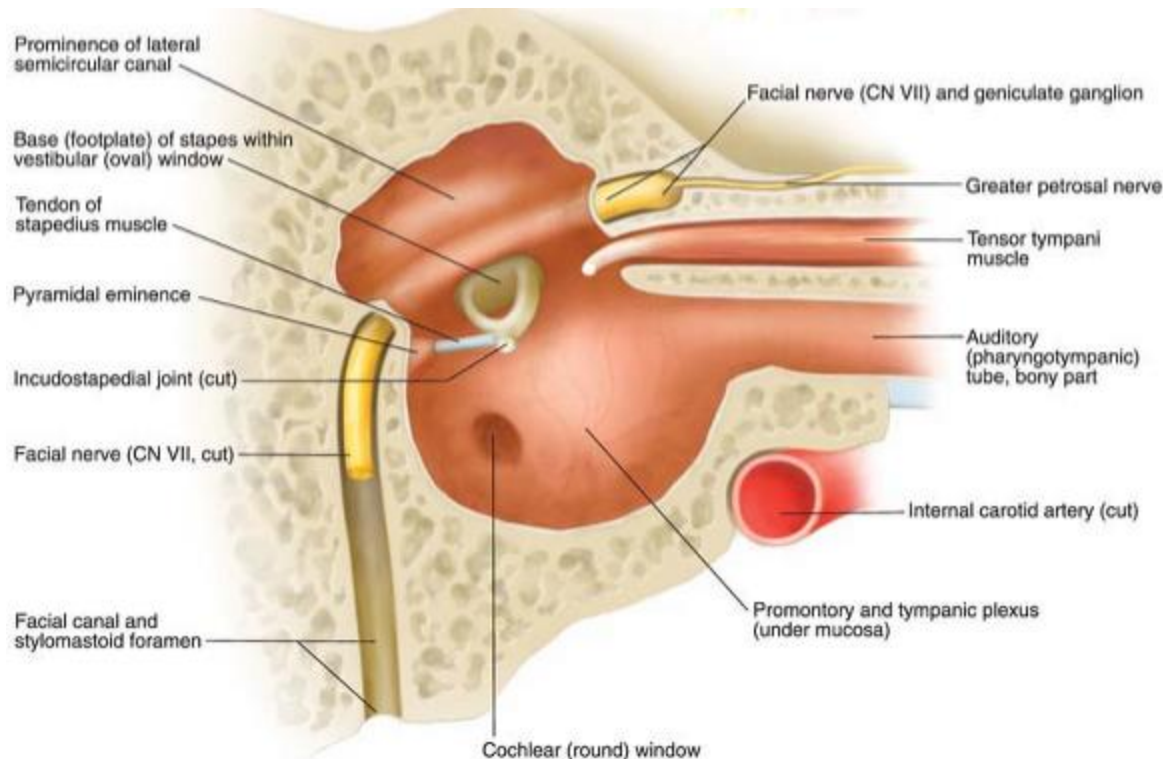


5. Anterior wall

- is only partially complete.
- The **lower part** consists of:
 - a thin layer of bone that separates the tympanic cavity from the internal carotid artery.
- **Superiorly:**
 - the wall is deficient due to the presence of a large opening for the entrance of the **pharyngotympanic tube** into the middle ear
 - a smaller opening for the canal containing the tensor tympani muscle.
- The foramen for the exit of the chorda tympani nerve from the middle ear is also associated with this wall

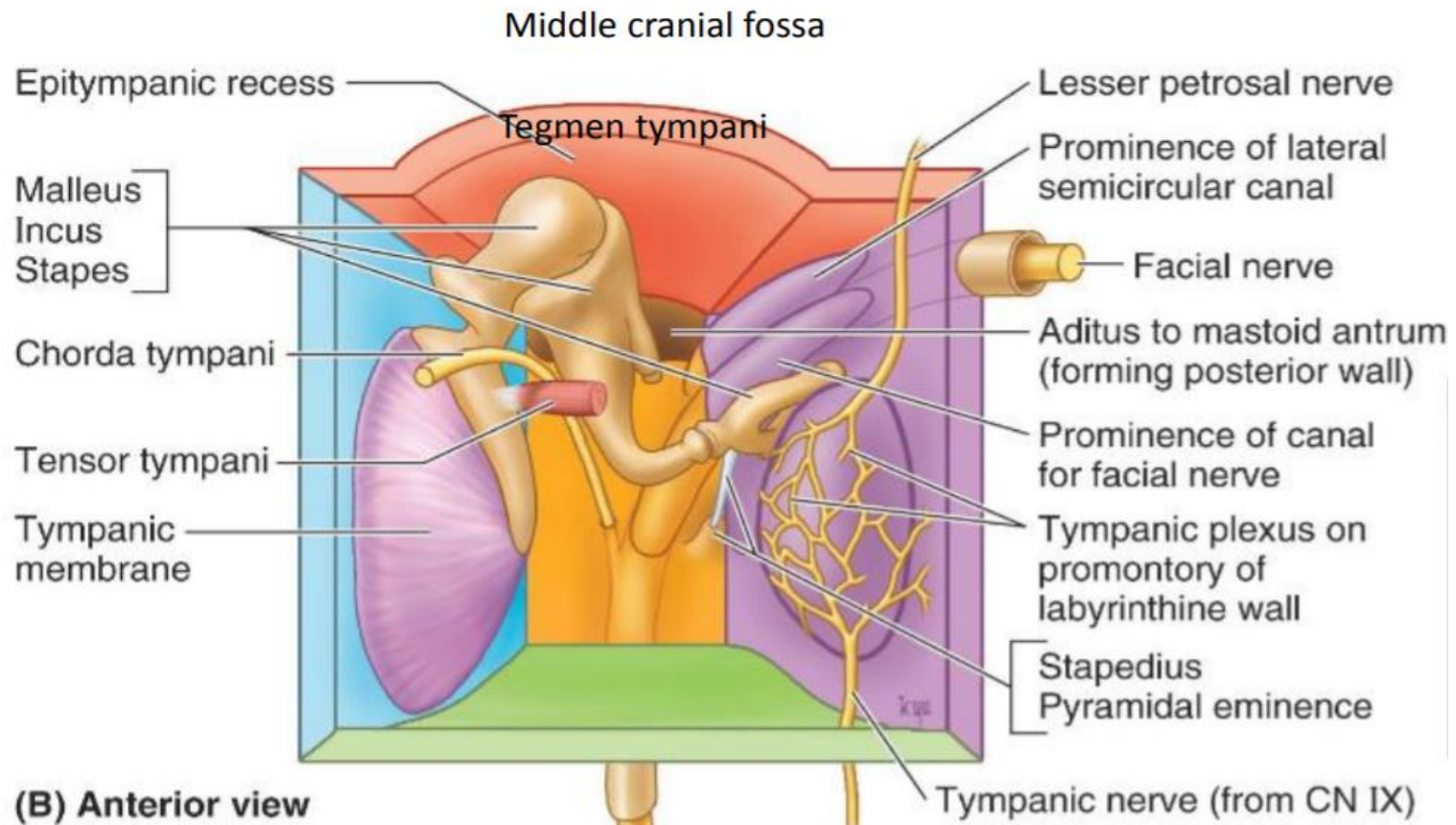
6. Labyrinthine wall (medial)

- is also the lateral wall of the internal ear
- A prominent structure on this wall is a **rounded bulge** (the **promontory**) produced by the basal coil of the cochlea
- Associated with the mucous membrane covering the promontory is a plexus of nerves (the **tympanic plexus**)

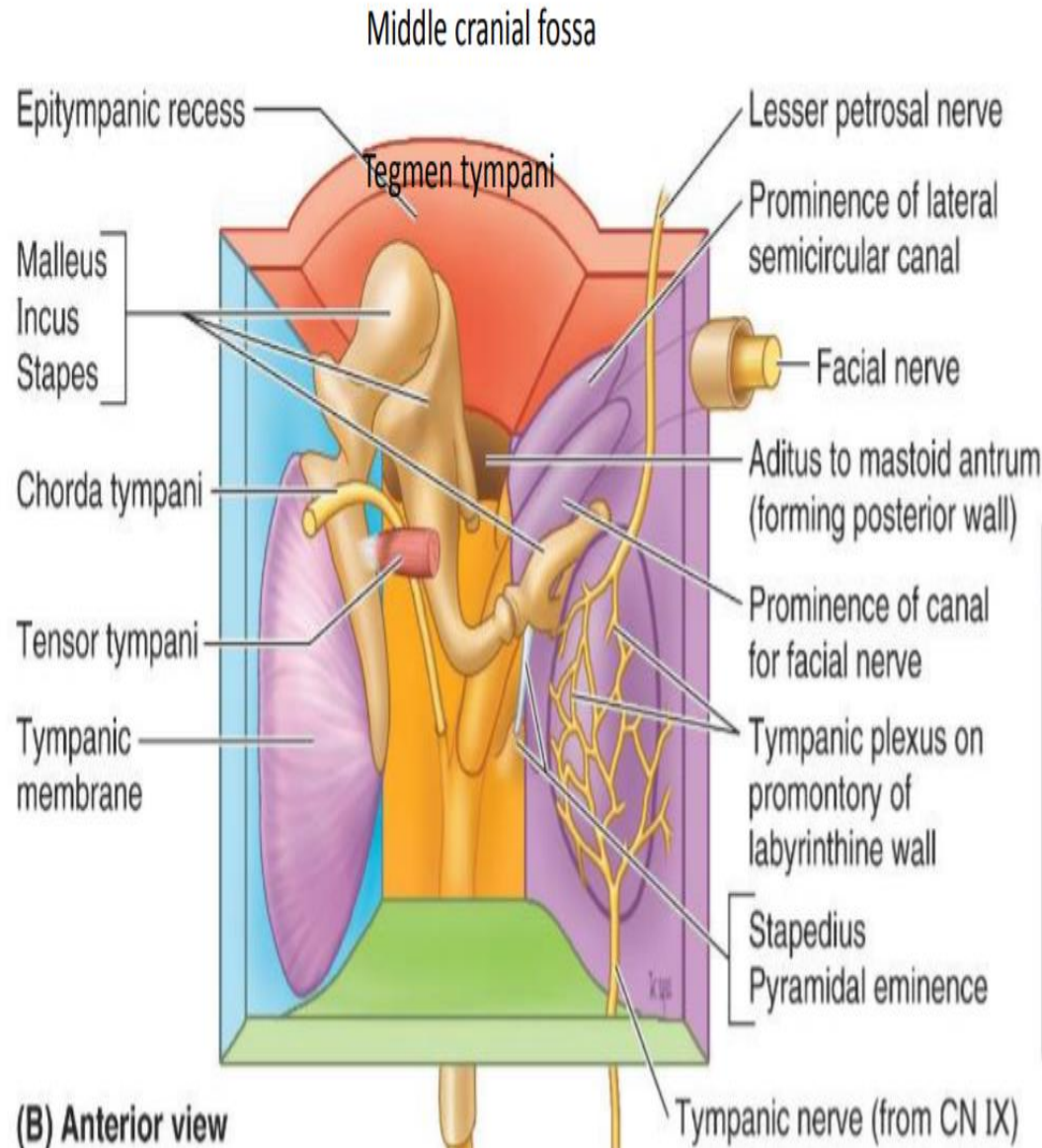


– the tympanic plexus:

- consists primarily of formed of :
 - the tympanic branch of **the glossopharyngeal nerve**
 - **branches from the internal carotid plexus.**
- Supplies:
 - the mucous membrane of the middle ear
 - the mastoid area
 - and the pharyngotympanic tube.



a branch of the tympanic plexus (**the lesser petrosal nerve**) leaves the promontory and the middle ear, travels across the anterior surface of the petrous part of the temporal bone, and leaves the middle cranial fossa through the **foramen ovale** to enter the otic ganglion

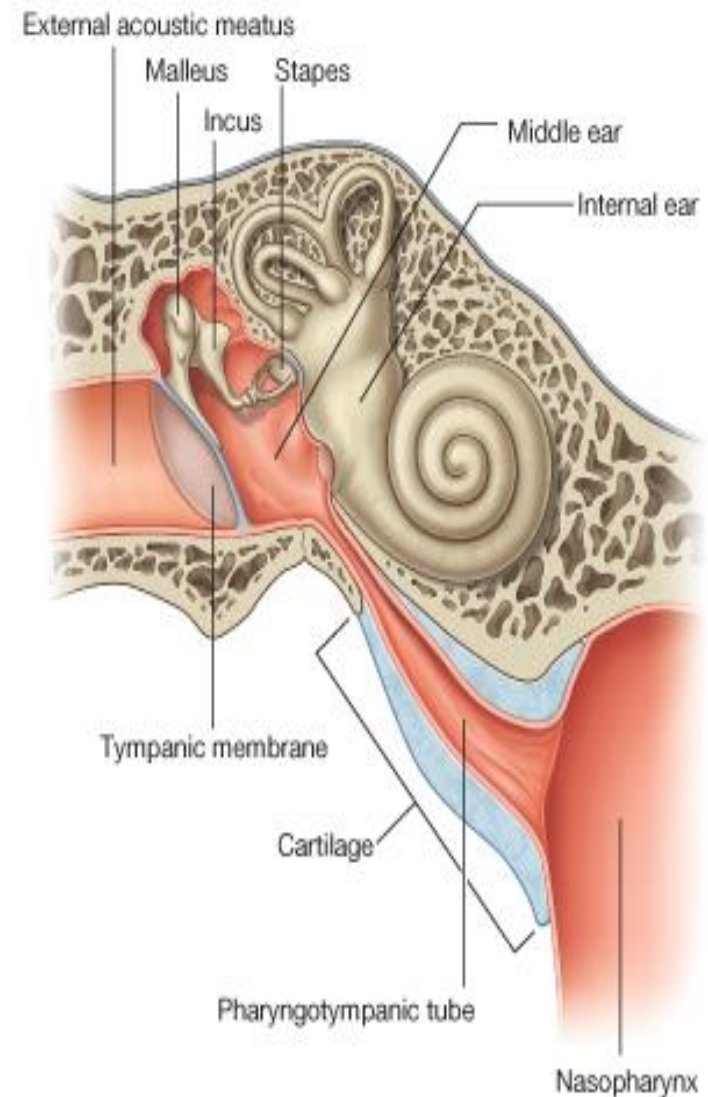


- The labyrinthine wall associated to two openings:
 - the oval and round windows
 - and two prominent elevations
- The oval window:
 - is located posterosuperior to the promontory
 - is the point of attachment for the base of stapes (footplate)
- The round window:
 - is posteroinferior to the promontory

- Posterior and **superior** to the **oval window** on the medial wall is **the prominence of facial canal**, which is a ridge of bone produced by the facial nerve [VII] in its canal as it passes through the temporal bone
- Just above and posterior to the prominence of facial canal is a broader ridge of bone (prominence of lateral semicircular canal) produced by the **lateral semicircular canal**, which is a structure involved in detecting motion.

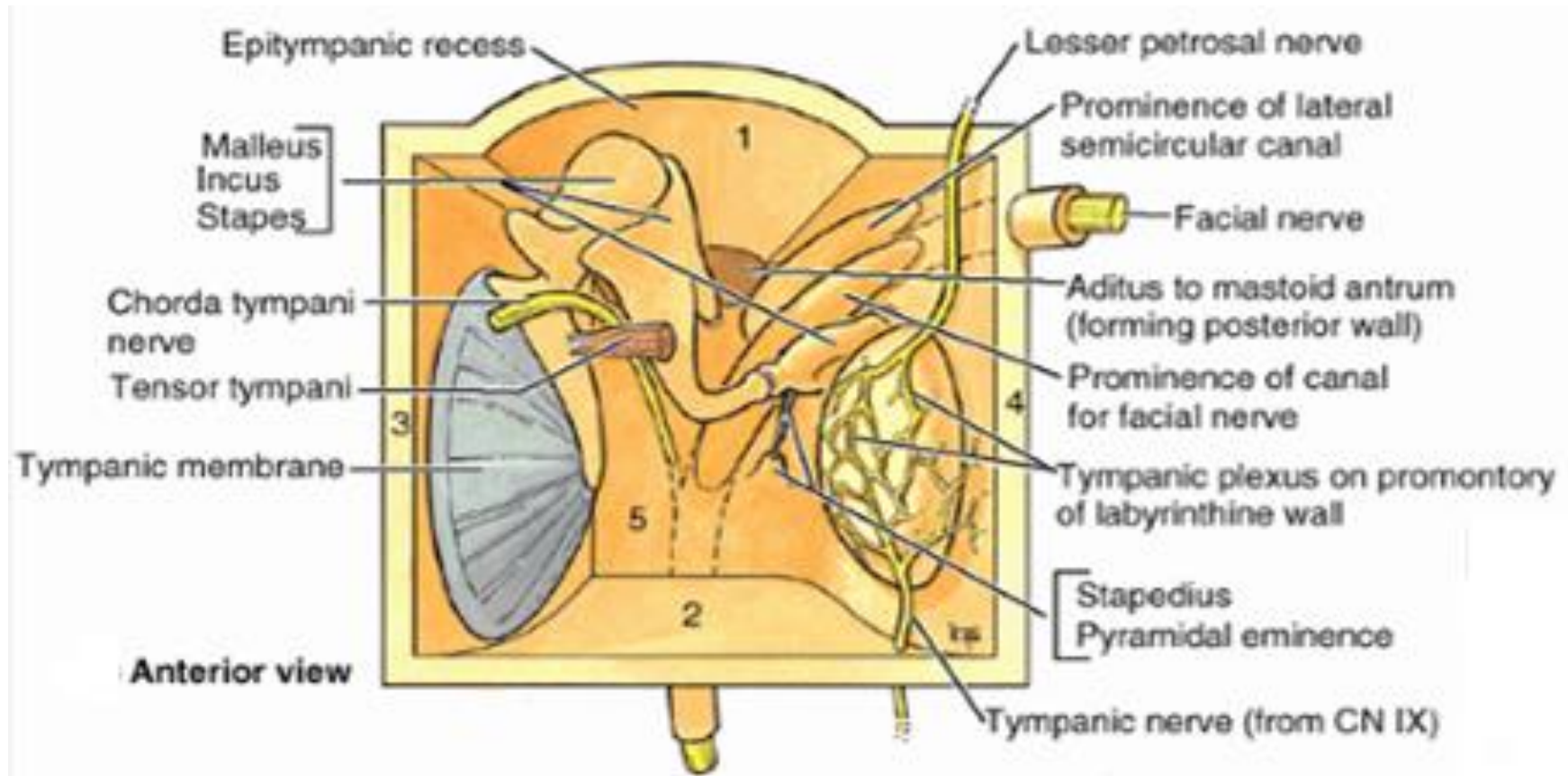
Pharyngotympanic tube

- connects the middle ear with the nasopharynx
- Used to equalizes pressure on both sides of the tympanic membrane
- Its opens into the middle ear is on the anterior wall
- and from here it extends forward, medially, and downward to enter the nasopharynx just posterior to the inferior meatus of the nasal cavity.
- It consists of:
 - **a bony part** (the one-third nearest the middle ear);
 - **a cartilaginous part** (the remaining two-thirds).



Contents of middle ear cavity

- Auditory ossicles (malleus, incus, and stapes).
- Stapedius and tensor tympani muscles.
- Chorda tympani nerve, a branch of CN VII
- Tympanic plexus of nerves



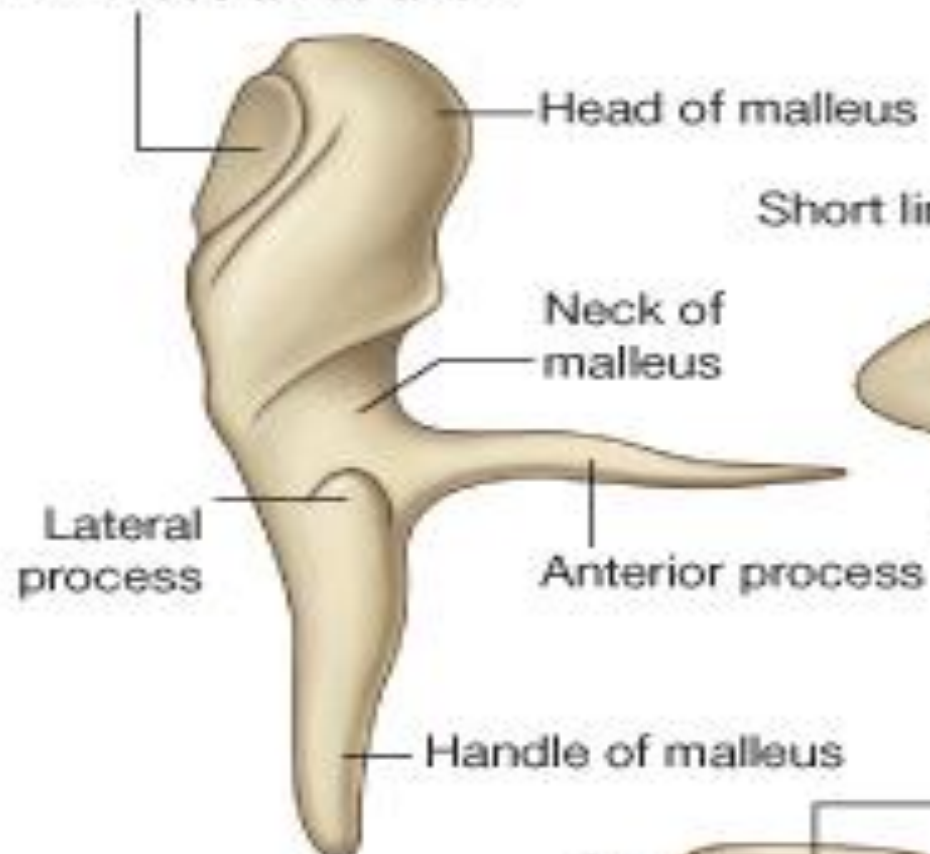
Auditory ossicles

- consist of the malleus, incus, and stapes.
- form an osseous chain across the middle ear from the tympanic membrane to the oval window of the internal ear
- Muscles associated with the auditory ossicles modulate movement during the transmission of vibrations.

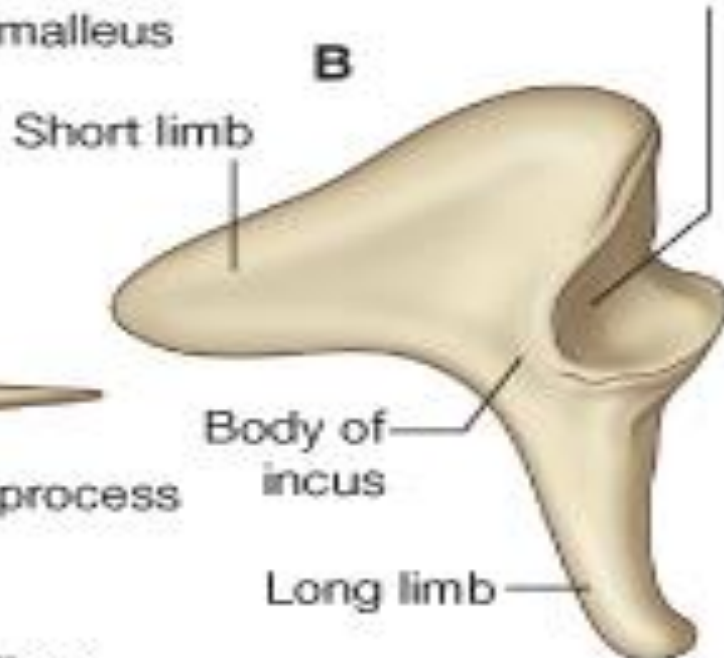
1. Malleus

- is the largest of the auditory ossicles
- is attached to the tympanic membrane.
- has the head, neck , anterior and lateral processes, and handle
- the anterior process is attached to the anterior wall of the middle ear by a ligament;
- the lateral process is attached to the anterior and posterior malleolar folds of the tympanic membrane.

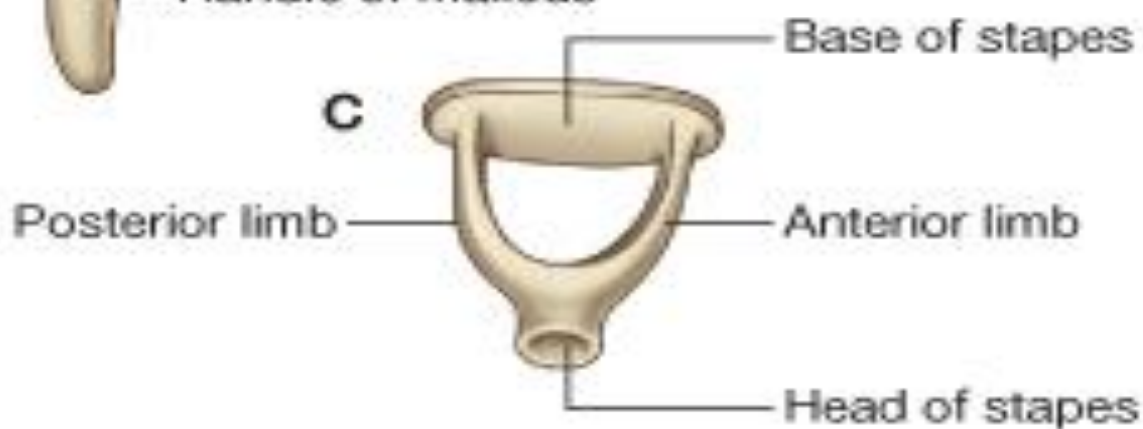
A Incus articulation



B Malleus articulation



C



2. Incus

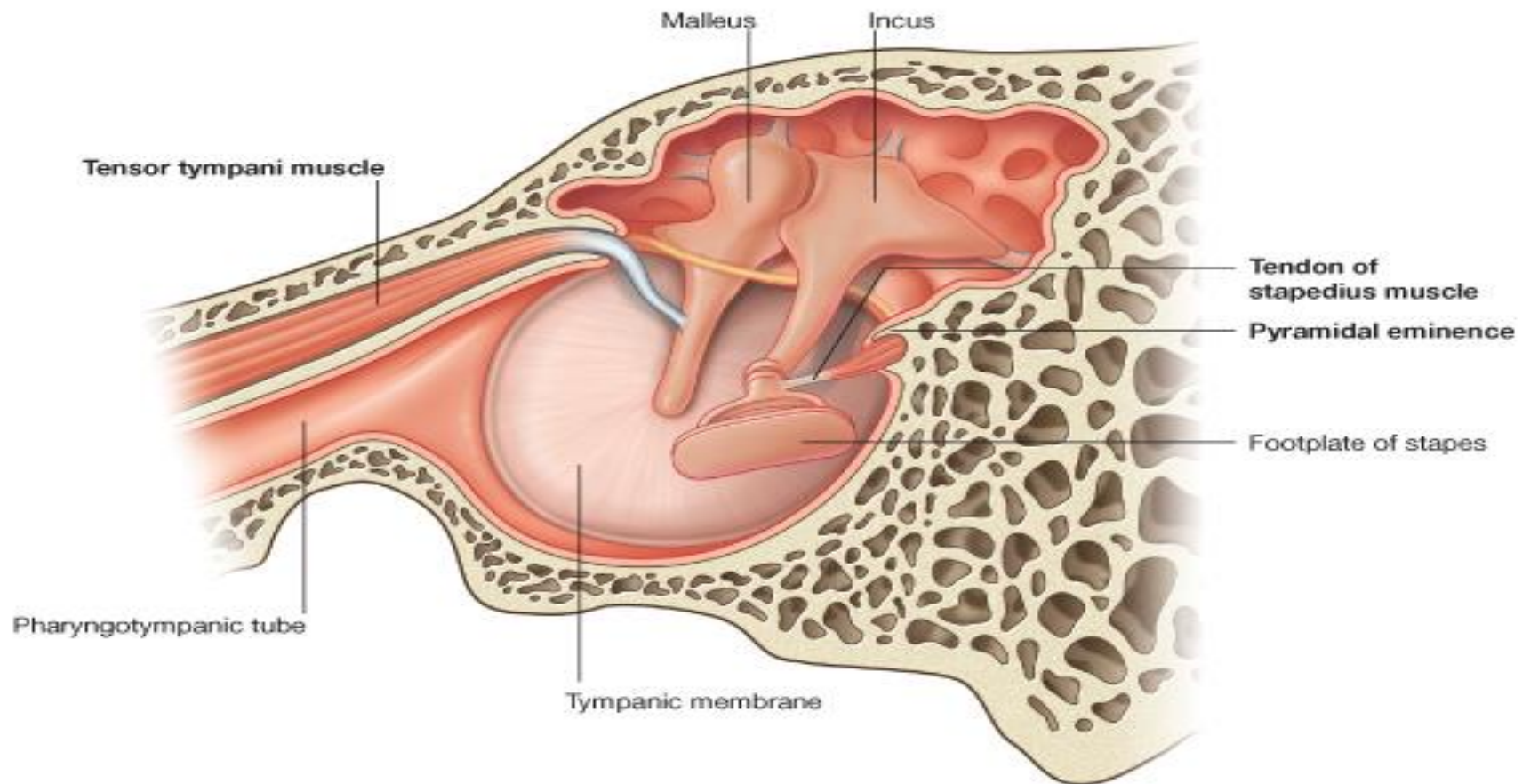
- Is the second bone in the series of auditory ossicles
- consists of the body, **long and short limbs**
- the enlarged body of incus articulates with the **head of malleus** and is in the **epitympanic recess**

3. Stapes

- is the most medial bone in the osseous chain
- is attached to the oval window.
- consists of the head, **anterior and posterior limbs**, and the base of stapes:
- the head of stapes is directed laterally and articulates with the **long process of the incus**
- the two limbs separate from each other and attach to the oval base;
- the base of stapes fits into the oval window on the labyrinthine wall of the middle ear.
-

Muscles of the Middle Ear

Muscle	Origin	Insertion	Nerve Supply	Action
Tensor tympani	Wall of auditory tube and wall of its own canal	Handle of malleus	Mandibular division of trigeminal nerve	Dampens down vibrations of tympanic membrane
Stapedius	Pyramid (bony projection on posterior wall of middle ear)	Neck of stapes	Facial nerve	Dampens down vibrations of stapes



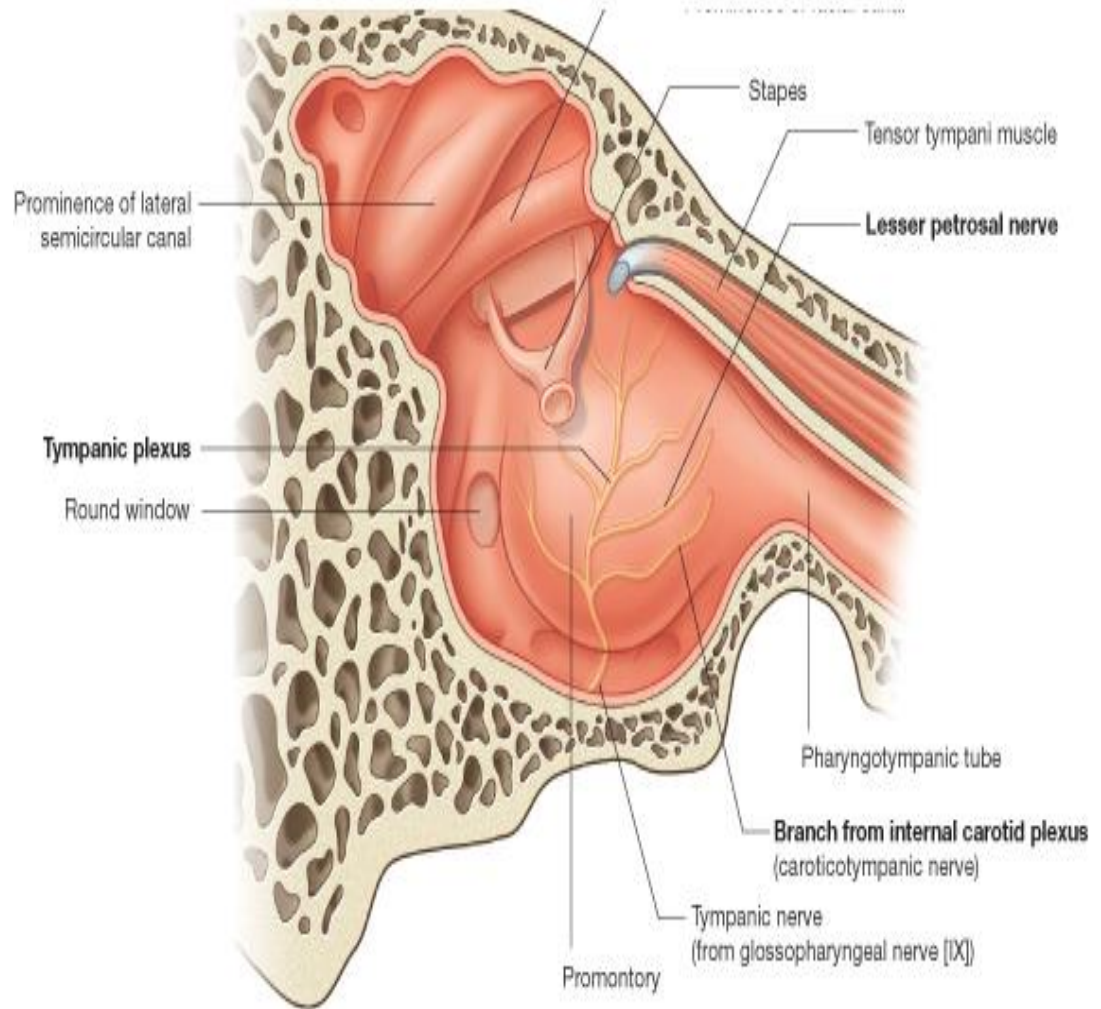
Vessels

- Arteries supply :
 - the two largest branches are:
 - the tympanic branch of the maxillary artery
 - the mastoid branch of the occipital
 - or posterior auricular arteries;
 - smaller branches come from:
 - the middle meningeal artery
 - the ascending pharyngeal artery
 - the artery of the pterygoid canal
 - and tympanic branches from the internal carotid artery
- Venous drainage:
 - returns to the pterygoid plexus of veins and the superior petrosal sinus

Innervation of the middle ear

- The tympanic plexus:
 - innervates the mucous membrane lining the walls
 - contents of the middle ear, which includes the mastoid area and the pharyngotympanic tube.
 - tympanic plexus is formed by:
 - the tympanic nerve, a branch of the glossopharyngeal nerve [IX]
 - from branches of the internal carotid plexus in the mucous membrane covering the promontory, which is the rounded bulge on the labyrinthine wall of the middle ear
- The glossopharyngeal nerve [IX]
 - exits the skull through the jugular foramen,
 - it gives off the tympanic nerve.
 - This branch re-enters the skull through a small foramen and passes through the bone to the middle ear.

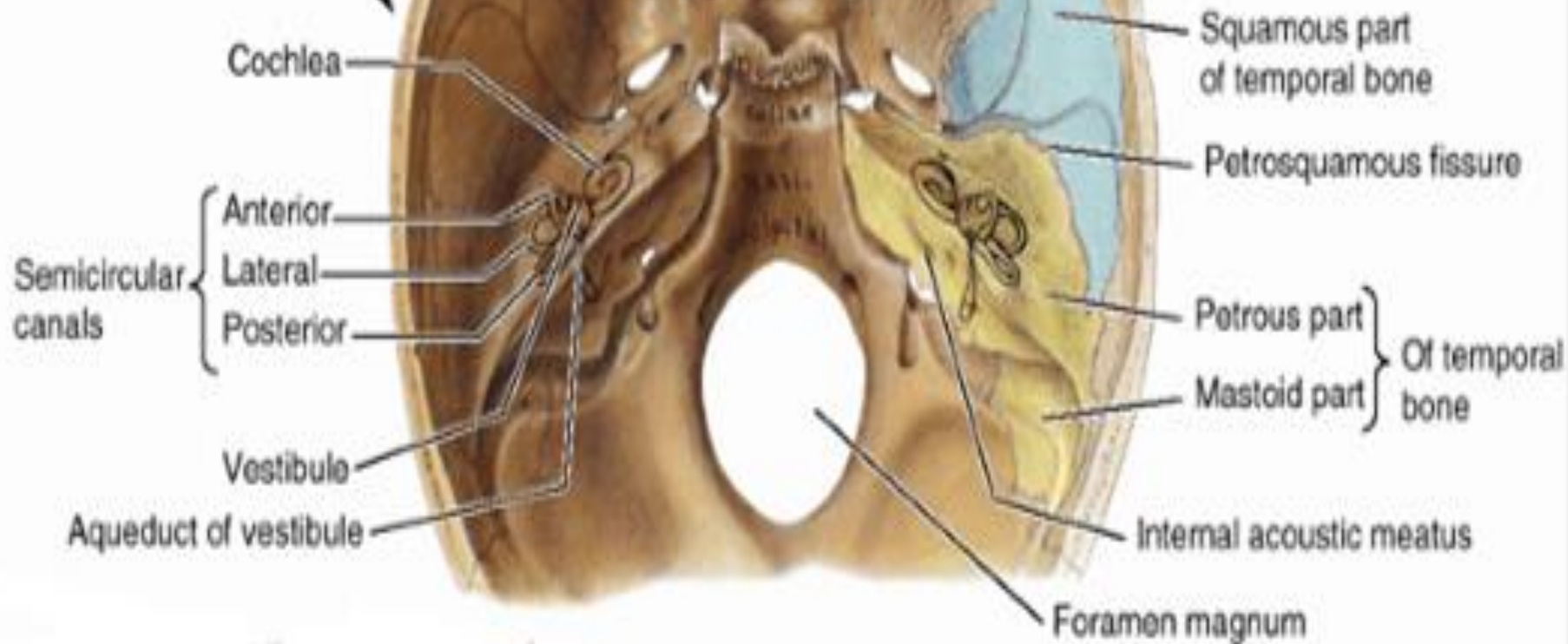
- Branches from the **tympanic plexus** supply the mucous membranes of the middle ear, including the pharyngotympanic tube and the mastoid area.
- The **tympanic plexus** also gives off a major branch (the lesser petrosal nerve), which supplies preganglionic parasympathetic fibers to the **otic ganglion**

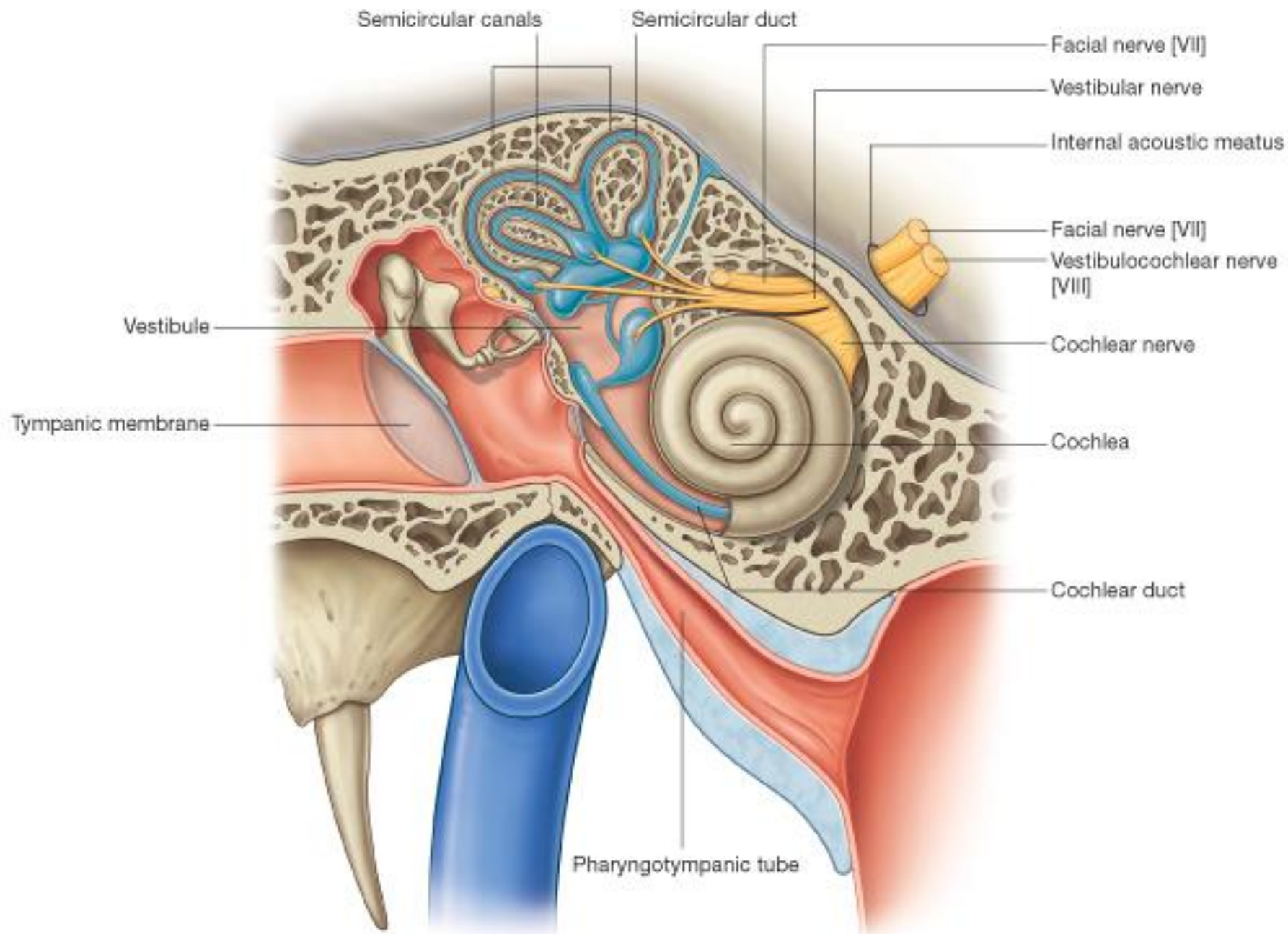


Internal ear

- consists of:
 - a series of bony cavities (**the bony labyrinth**)
 - and membranous ducts and sacs (the **membranous labyrinth**) within these cavities.
 - All these structures are in the petrous part of the temporal bone between the **middle ear laterally** and the **internal acoustic meatus medially**
- The bony labyrinth:
 - ✓ consists of the vestibule
 - ✓ three semicircular canals
 - ✓ and the cochlea.
 - ✓ These bony cavities are lined with periosteum and contain a clear fluid (**the perilymph**).

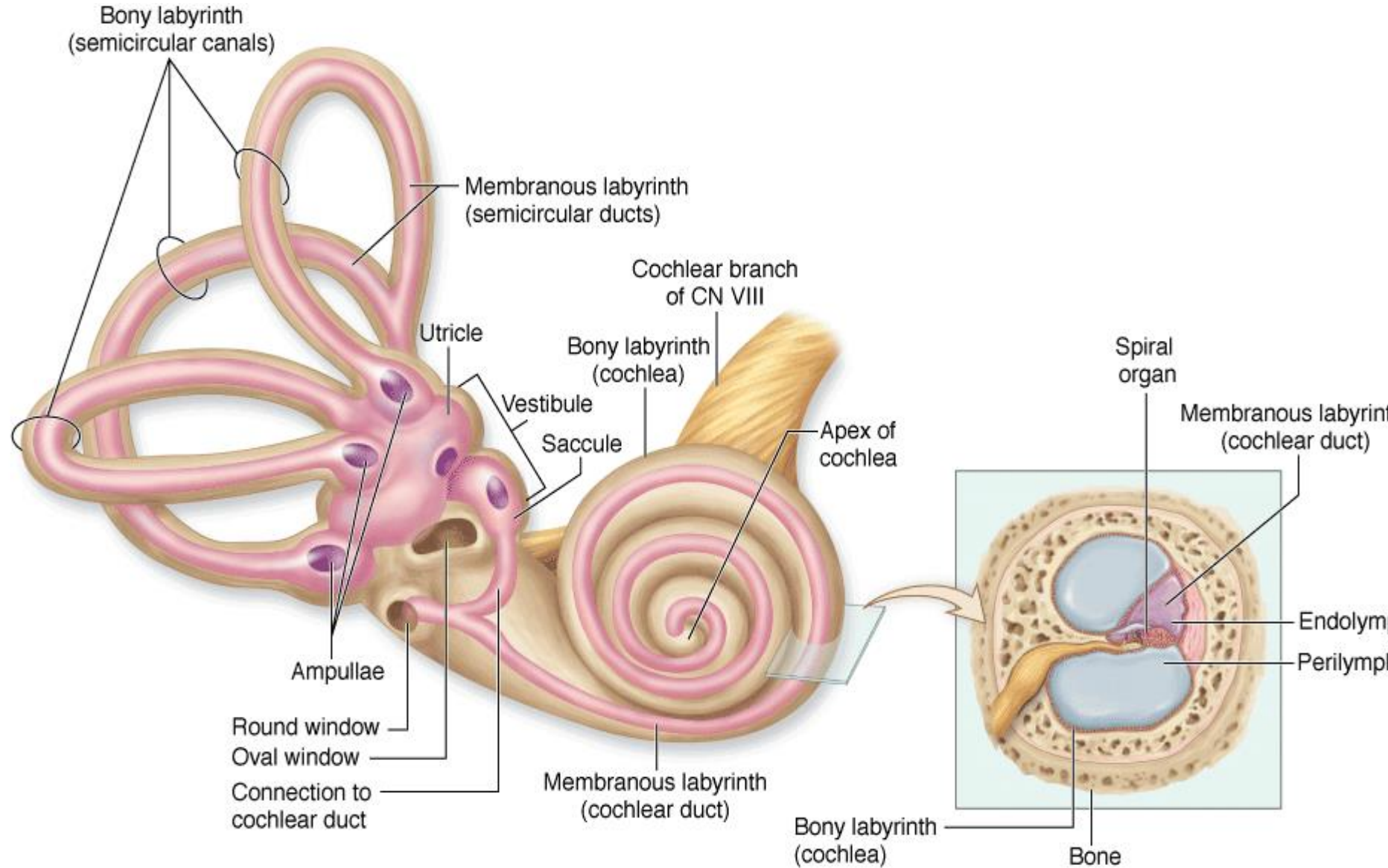
Direction of view for
parts B-D





Bony Labyrinth

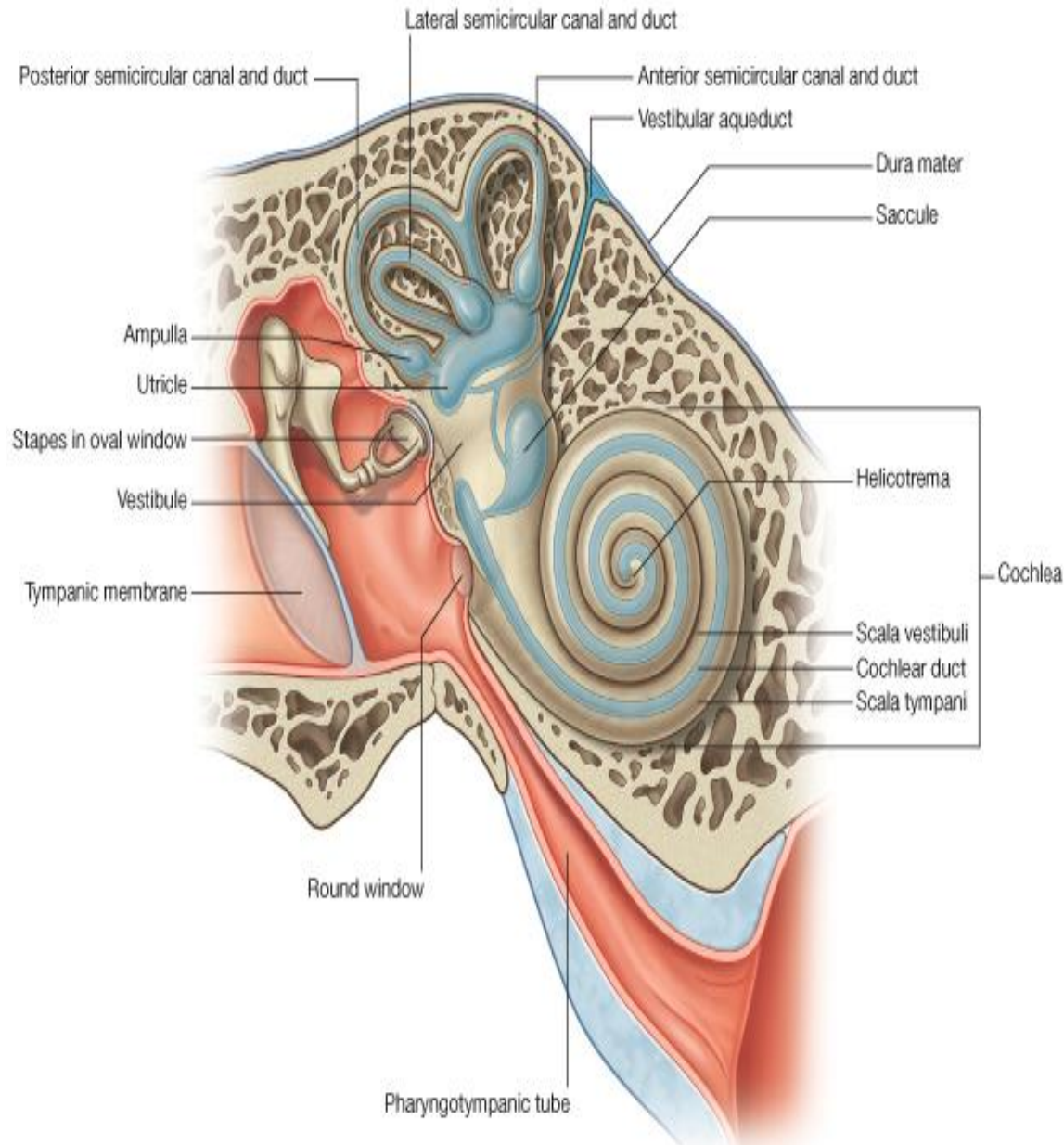
- is a series of cavities (cochlea, vestibule, and semicircular canals) contained within the **otic capsule of the petrous** part of the temporal bone
- The **otic capsule** is made of bone that is denser than the remainder of the petrous temporal bone
- The **otic capsule** is often erroneously illustrated and identified as being the bony labyrinth.
- The bony labyrinth is **the fluid-filled space**
- communicates with the **subarachnoid space** superior to the jugular foramen through the **cochlear aqueduct**
- It also features the round window (L. fenestra cochleae), closed by the secondary tympanic membrane



The internal region of the ear is composed of a cavity in the temporal bone, the bony labyrinth, which houses a fluid-filled membranous labyrinth. The membranous labyrinth includes the vestibular organs for the sense of equilibrium and balance (the saccule, utricle, and semicircular ducts) and the cochlea for the sense of hearing.

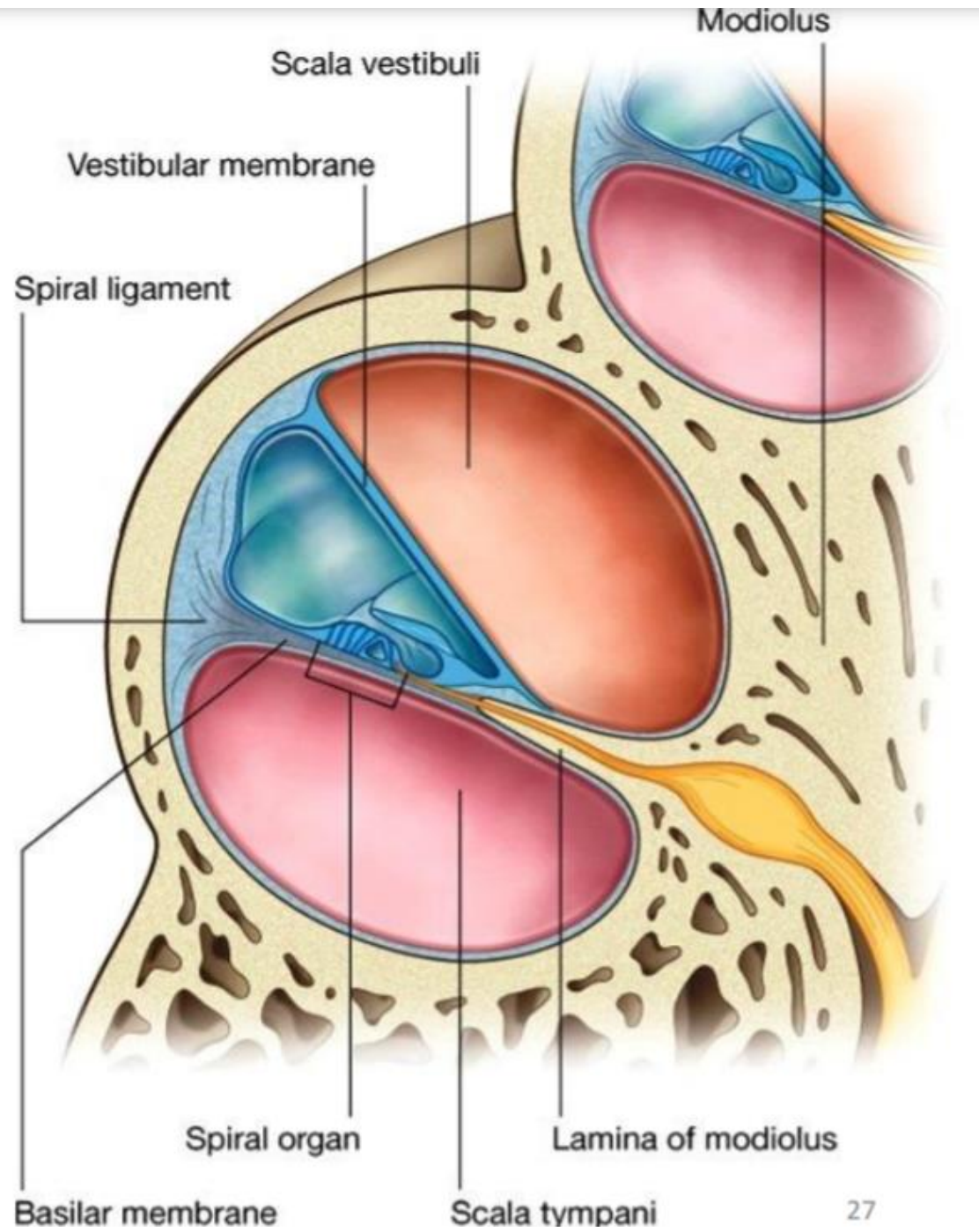
Cochlea

- is the shell-shaped part of the bony labyrinth that contains the **cochlear duct**
- the part of the internal ear concerned with hearing.
- The spiral canal of the cochlea begins at the vestibule and makes 2.5 turns around a bony core, **the modiolus**
- The modiolus contains canals for blood vessels and for distribution of the branches of the cochlear nerve



Cochlear Duct

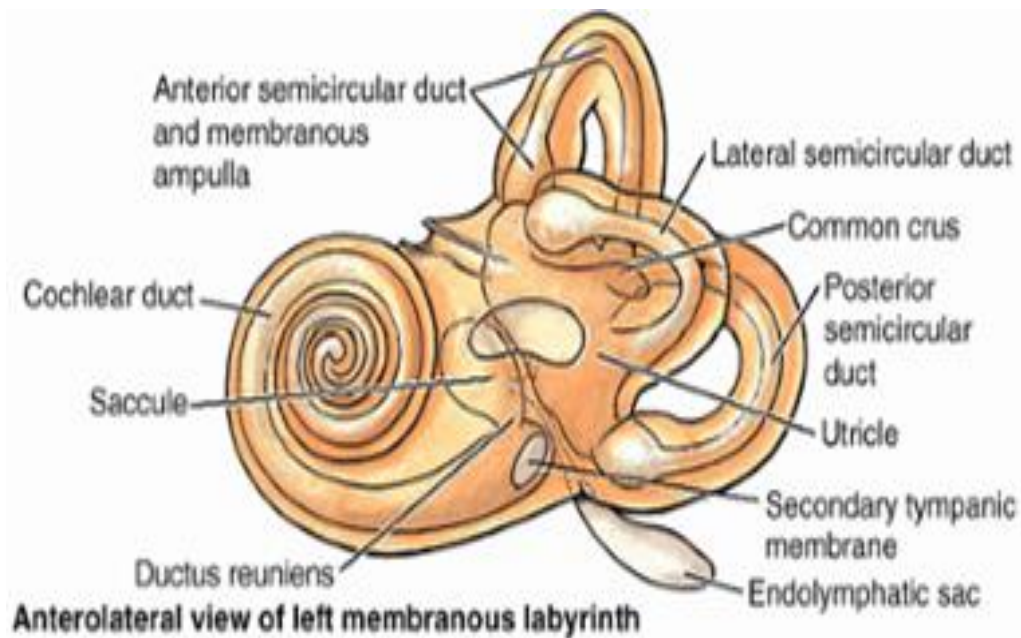
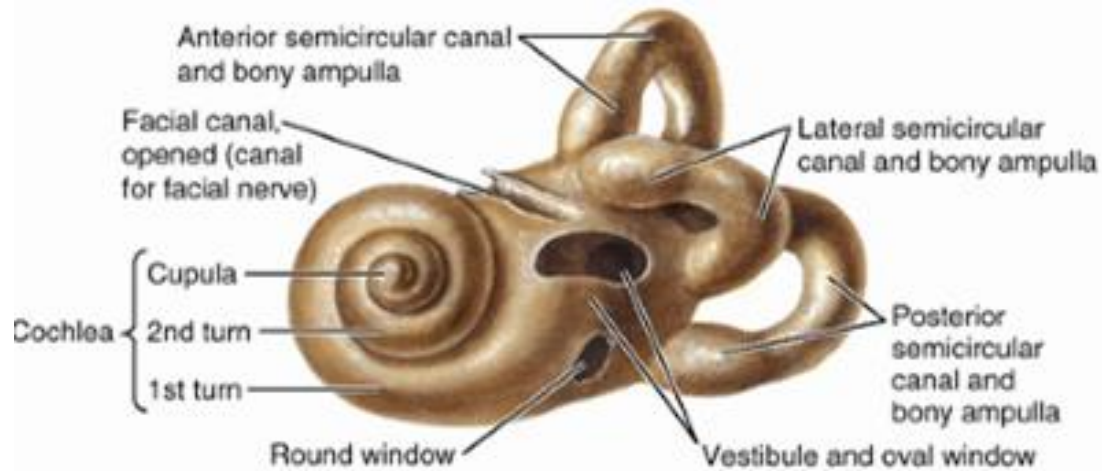
- Contains endolymph
- Lies between scala tympani and scala vestibuli
- Pressure waves travel up and down the scala and vibrate the basilar membrane
- Organ of Corti (spiral organ) sits on basilar membrane



Vestibule of the Bony Labyrinth

- is a small oval chamber (approximately 5 mm long)
- contains the **utricle** and **saccul**
- and parts of the balancing apparatus (vestibular labyrinth).
- The vestibule features **the oval window** on its lateral wall, occupied by the base of the stapes.
- The aqueduct extends to the posterior surface of the petrous part of the temporal bone, where it opens posterolateral to the internal acoustic meatus
- The vestibular aqueduct transmits the endolymphatic duct and two small blood vessels poked

Anterolateral view of left otic capsule

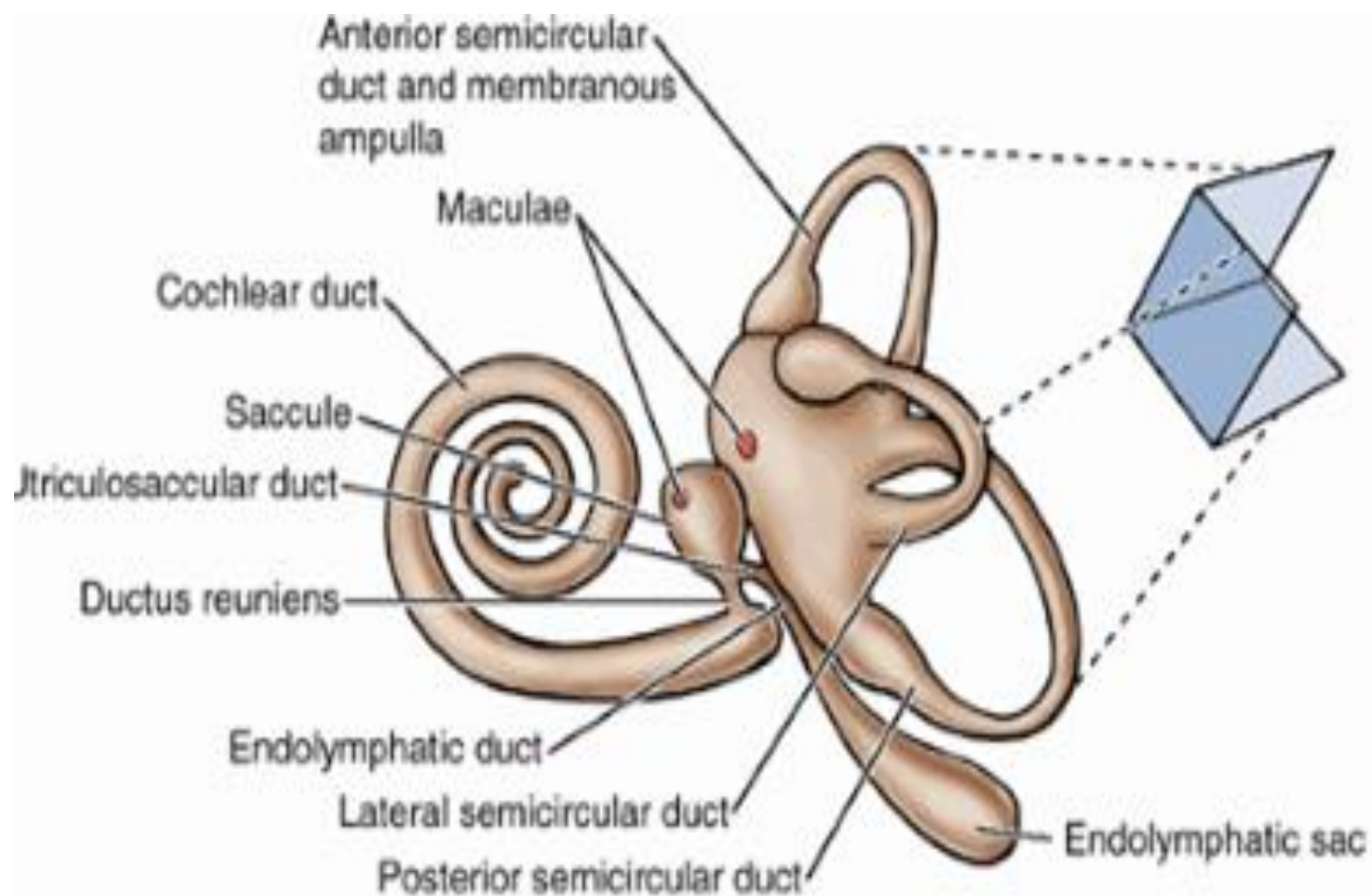


Semicircular Canals

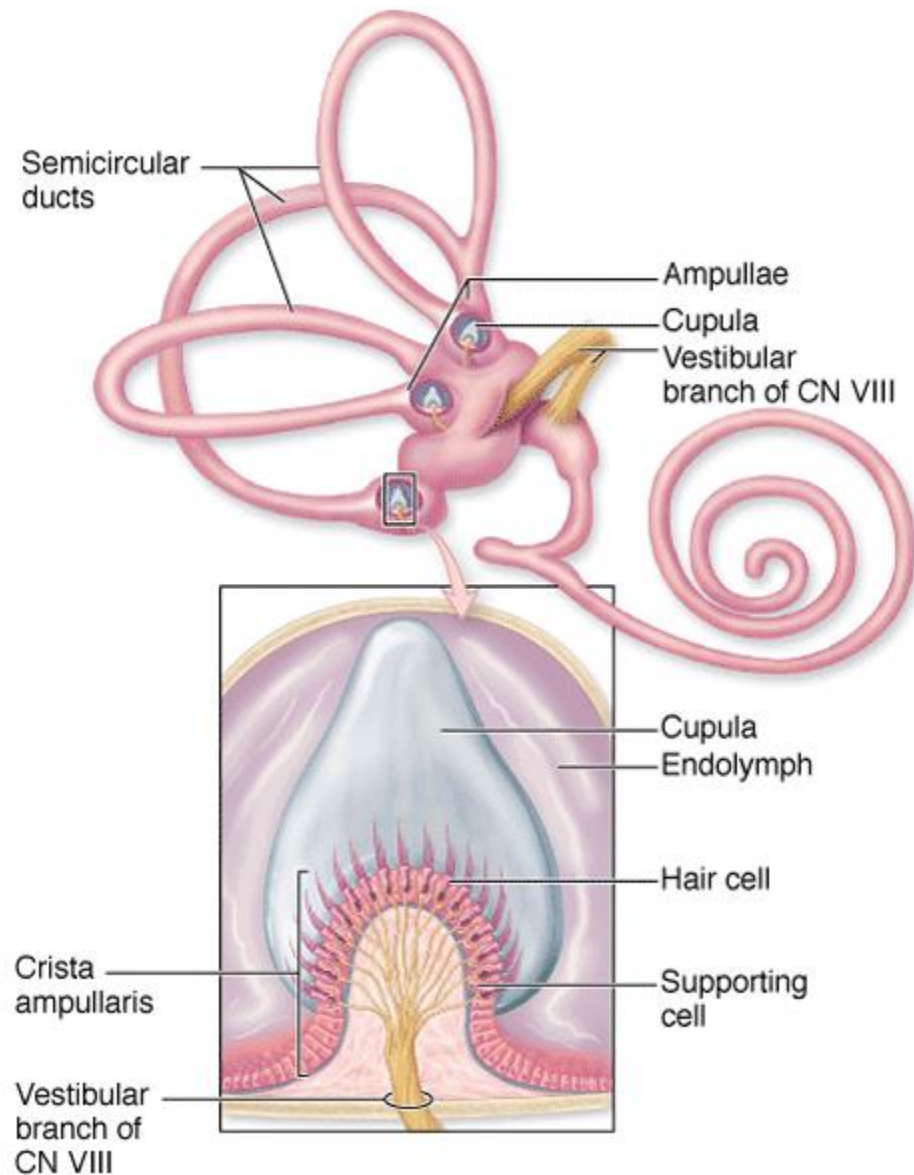
- Consists of anterior, posterior, and lateral communicate with the vestibule of the bony labyrinth
- The canals lie posterosuperior to the vestibule into which they open
- they are set at right angles to each other.
- The canals occupy three planes in space.
- Each semicircular canal forms approximately two thirds of a circle and is approximately **1.5 mm in diameter**, except at one end where there is a swelling, the bony ampulla.
- The canals have only five openings into the vestibule because the anterior and posterior canals have one limb common to both.
- Lodged within the canals are the **semicircular ducts**

Semicircular Ducts (three)

- parts of the membranous labyrinth having the same general form as **the semicircular canals** in the bony labyrinth
- Each extends from and returns to the wall of the utricle
- lie in three different spatial planes, at approximately right angles to one another
- The enlarged ampulla end of each semicircular duct has an elongated ridgelike area of mechanoreceptors called the crista ampullaris
- The ridge of each crista ampullaris is perpendicular to the long axis of the duct.

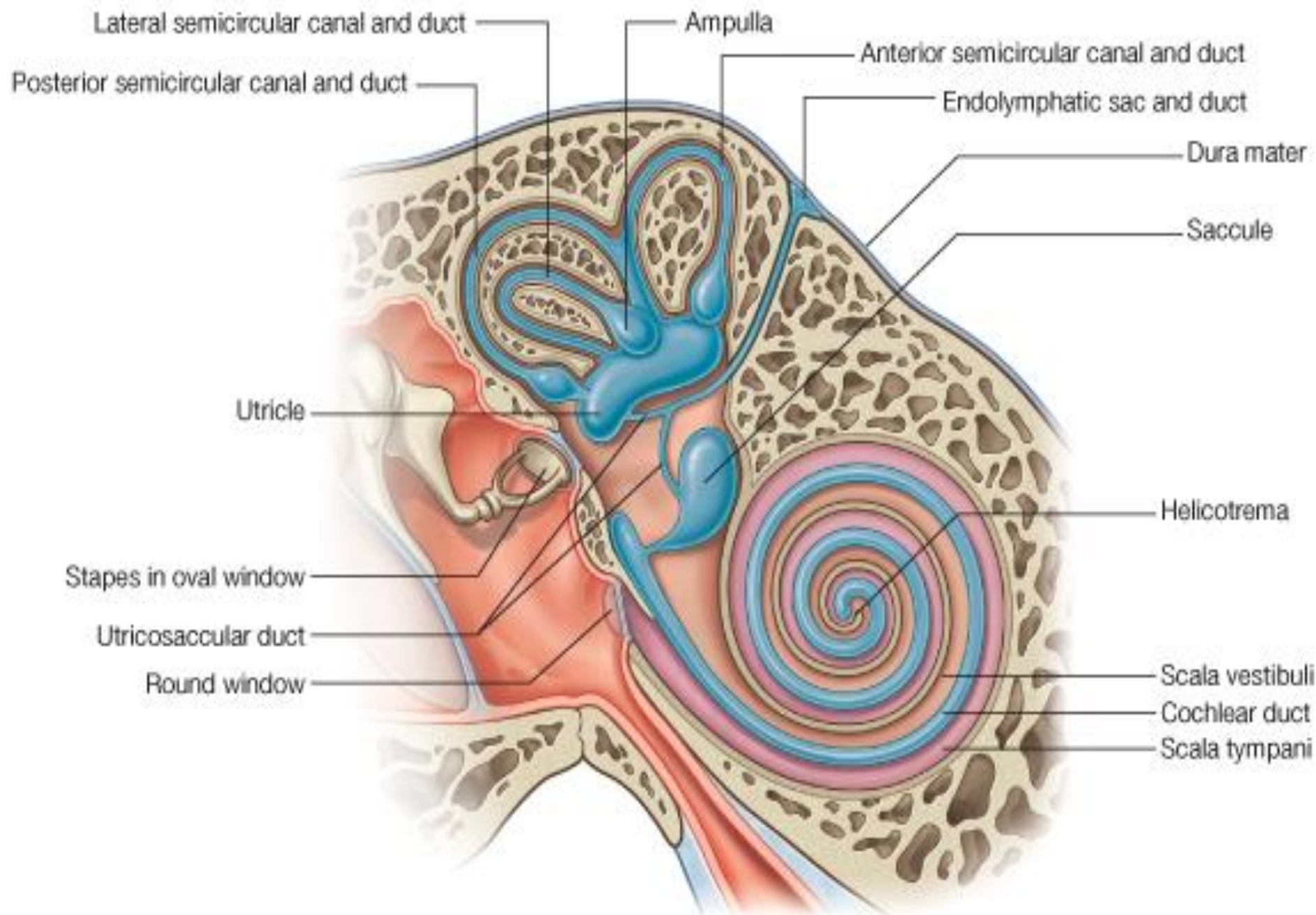


- Each of the semicircular ducts has an expanded end called the ampulla
- The wall of each ampulla is raised as a ridge called the **crista ampullaris**, a section of which is shown here diagrammatically.
- Hair cells of the crista epithelium resemble the two types found in the maculae, with hair bundles projecting into a dome-shape overlying layer of proteoglycan called the cupula
- The cupula is attached to the wall opposite the crista and is moved by endolymph movement within the semicircular duct.



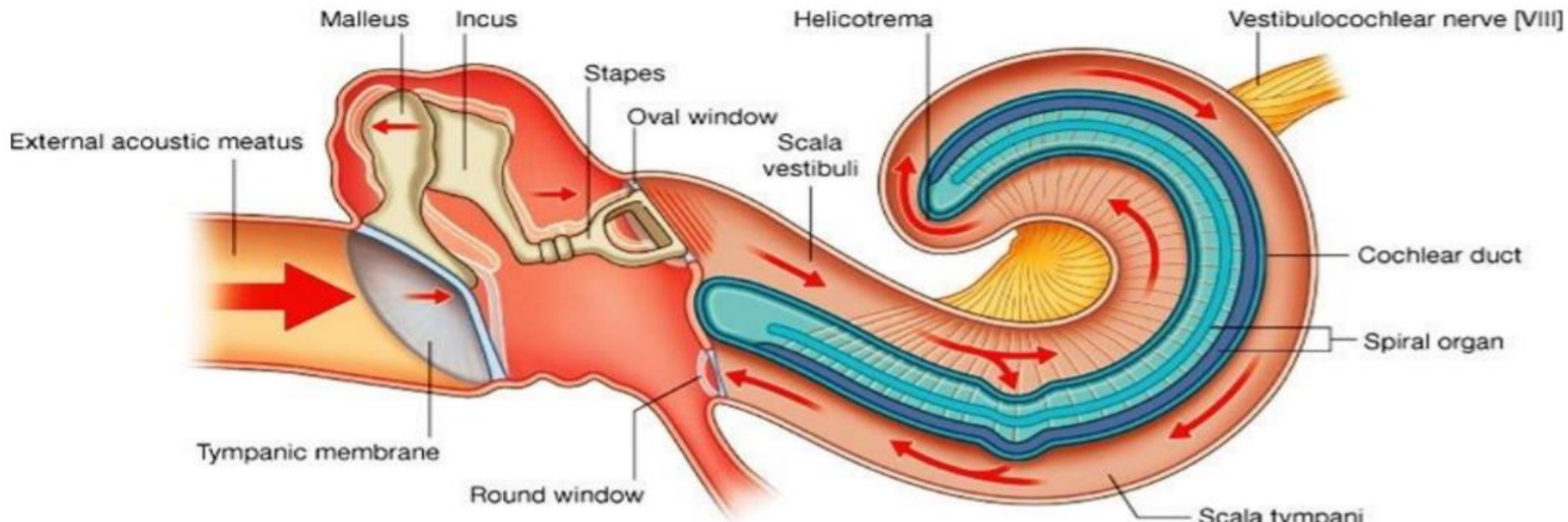
Membranous Labyrinth

- consists of a series of **communicating sacs** and ducts that are suspended in the bony labyrinth
- **contains endolymph, a watery fluid similar** in composition to intracellular fluid, thus differing in composition from the surrounding perilymph (which is like extracellular fluid) that fills the remainder of the bony labyrinth.
- **composed of two divisions:**
 - the vestibular labyrinth and the cochlear labyrinth
 - consists of more parts than does the bony labyrinth:
- **Vestibular labyrinth:**
 - utricle and saccule
 - two small communicating sacs in the vestibule of the bony labyrinth.
 - Three semicircular ducts in the semicircular canals.
- **Cochlear labyrinth:** cochlear duct in the cochlea.



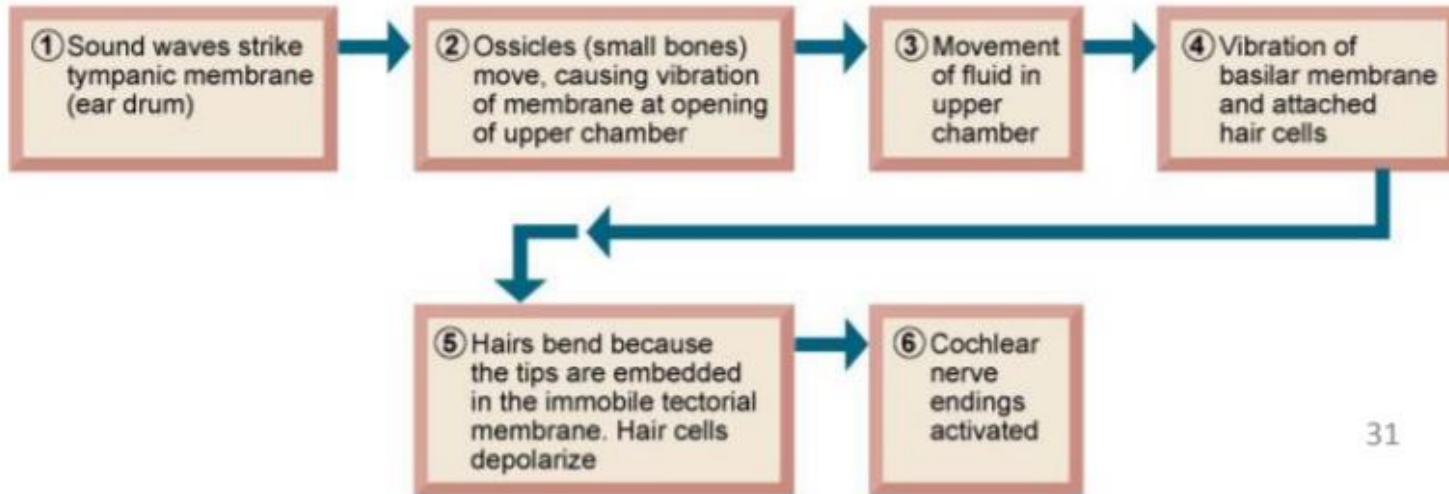
Transmission of Sound

- Stapes moves in and out at oval window
- Sound waves transmitted up scala vestibuli toward helicotrema
- Frequency of sound wave causes specific portion of basilar membrane to vibrate
- Hair cells at that location create action potentials
- Pressure wave travels down scala tympani and pressure released at round window



Middle Ear

Inner Ear



- **Reading references:**

- 1.**

- 2.**

THANKS